Introduction & Classification of Fatty Acids & Lipids

Program: BDS (1st year)
Date: 20/ 06/2016

LECTURE OBJECTIVES

- At the end of lecture the student should be able to:
  - Define fatty acids
  - Describe the structure of fatty acids
  - Classify fatty acids on the basis of saturation, shape and nutrition
  - Define lipids
  - Classify lipids
  - Describe the functions of various classes of lipids.

FATTY ACIDS

- Mono carboxylic acids with a hydrocarbon chain of varying length

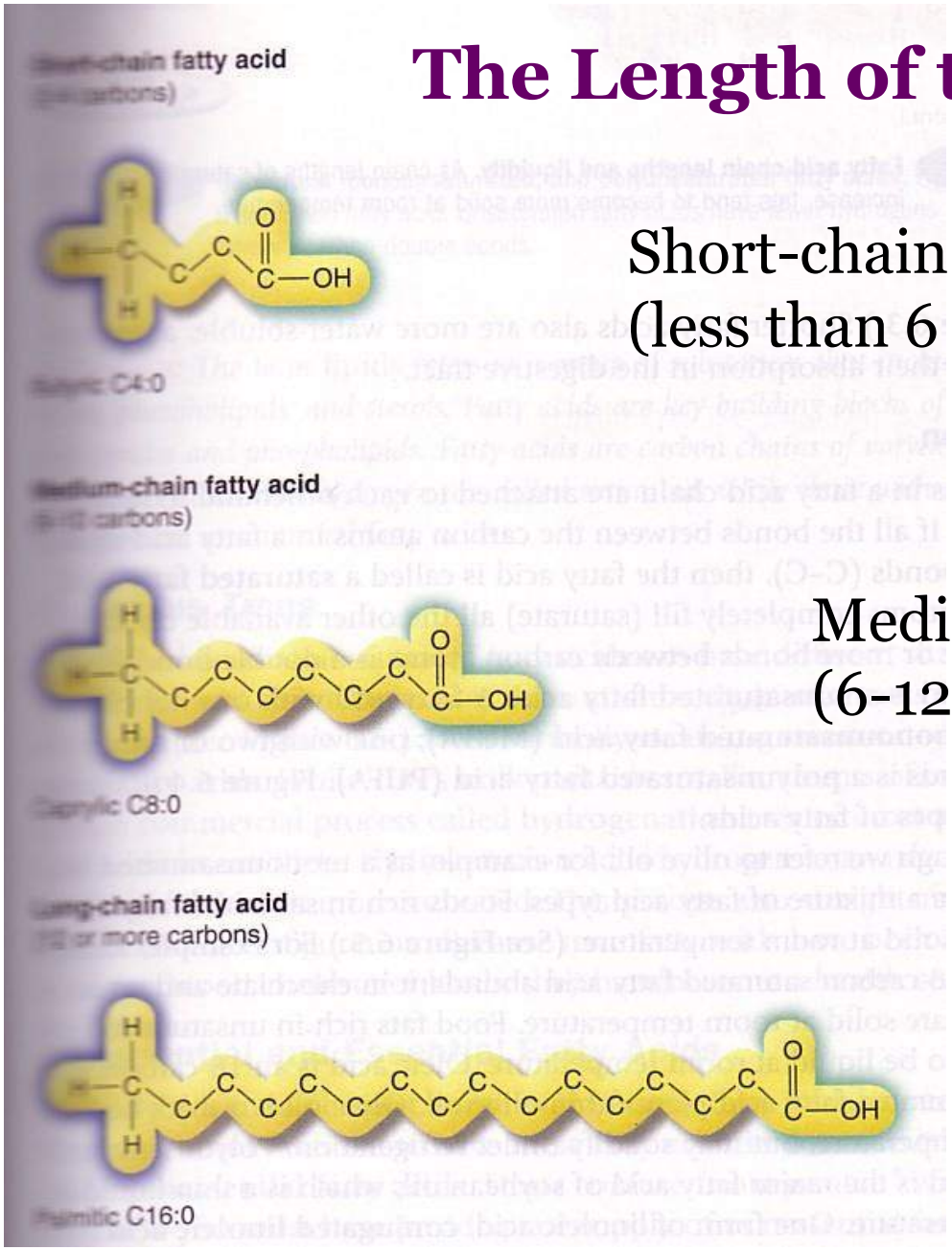


- Hydrocarbon chain with two ends
  - Carboxylic acid
  - Methyl group

Fatty Acids

- Fatty acids are aliphatic carboxylic acids
  - Have the general formula R-$(CH2)_n$-COOH
- Fatty acids in biological systems usually contain an even number of carbon atoms,
  - typically between 14 and 24.
  - The 16- and 18-carbon fatty acids are most common.
- They occur mainly as esters in natural fats & oils
  - but do occur in the unesterified form as free fatty acids, a transport form found in the plasma.

The Length of the Carbon Chain



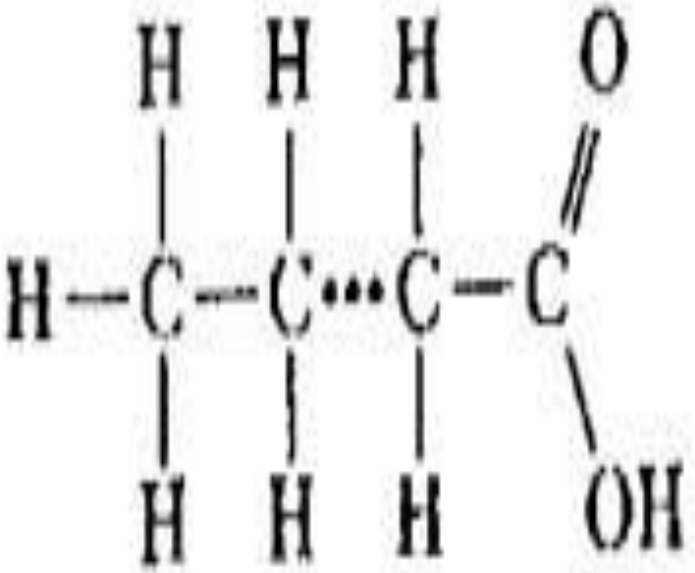
Short-chain Fatty Acid
(less than 6 carbons)

Medium-chain Fatty Acid
(6-12 carbons)

Long-chain Fatty Acid
(12 or more carbons)

Structure of fatty acids

- A fatty acid is nothing more than a long C-H chain with a carboxyl group (COOH) on the end.
  - The COOH gives it an acid property.
  - The 3….dots represent the chain is very long.



Fatty Acids: Structure

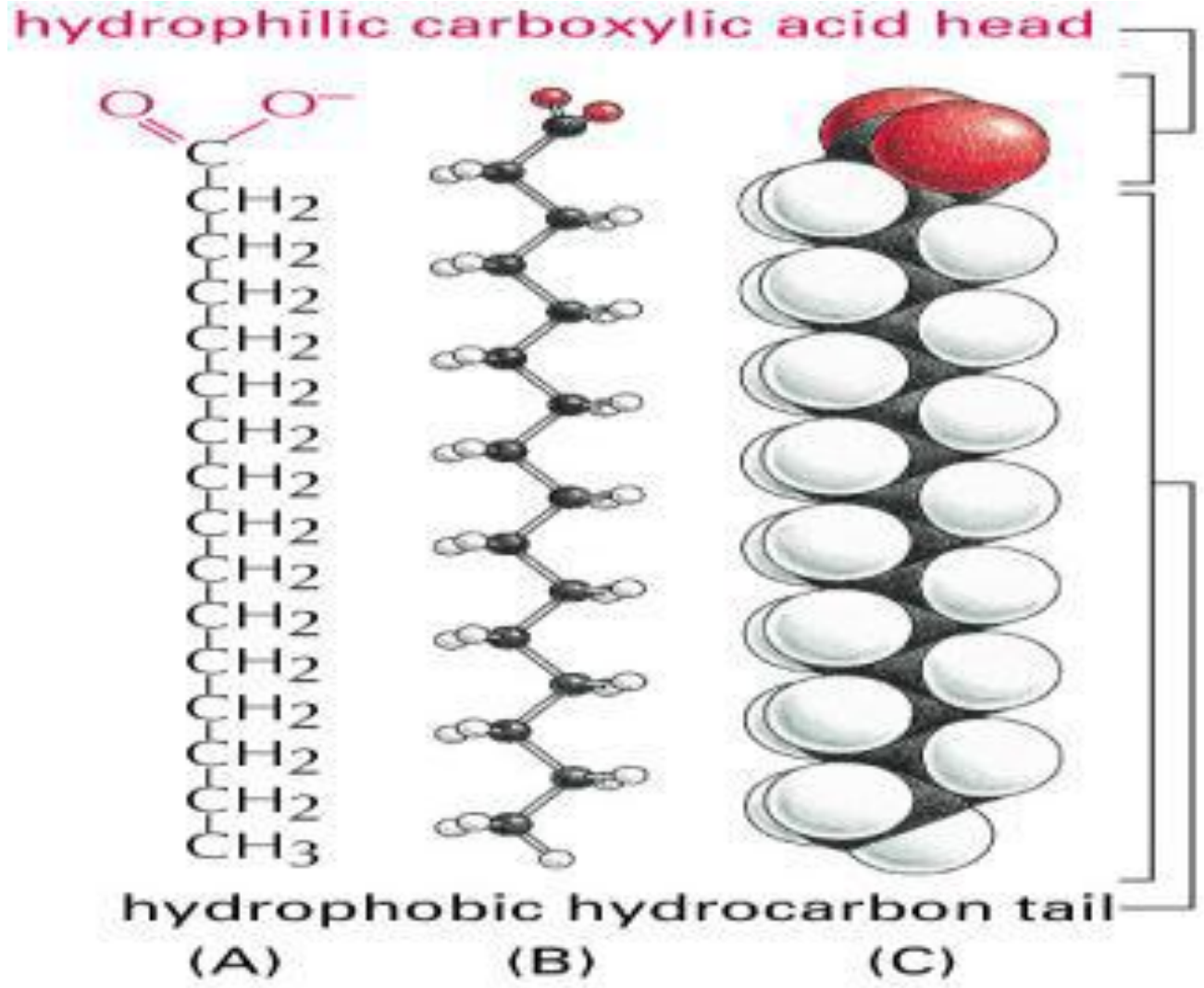


Figure 2–21. Molecular Biology of the Cell, 4th Edition.

Fatty Acids Classification

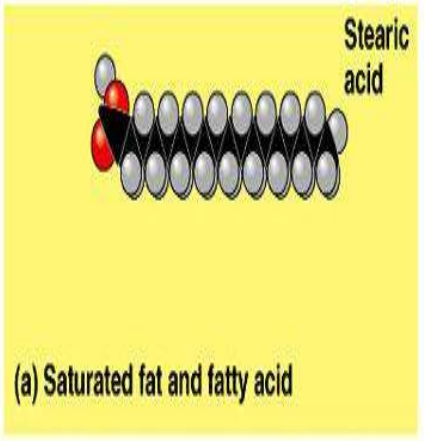
- Double bond
  - Saturated fatty acids
  - Unsaturated fatty acids
- Shape
  - Straight (Trans configuration)
  - Kinked (Cis Configuration)
- Nutrition
  - Essential fatty acids
  - Non essential fatty acids

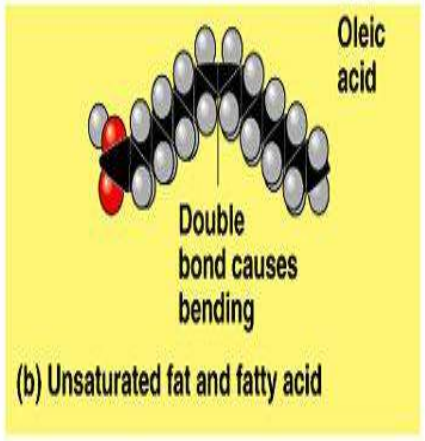
Saturated and Unsaturated FAs

- Saturated FAs have no double bonds.
  - (C-C)
- Unsaturated FAs have at least one double bond (C=C)
- Divided into following categories
  - Monounsaturated, only one double bond.
  - Polyunsaturated – many double bonds.









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Saturated Fatty Acid

- All single bonds between carbons



Saturated Fatty Acids

Number of C atoms	Common Name	Systemic Name	Formula
2	Acetic acid	Ethanoic acid	CH3COOH
4	Butyric acid	Butanoic acid	$CH_3(CH_2)_2COOH$
6	Caproic acid	Hexanoic acid	$CH_3(CH_2)_4COOH$
8	Caprylic acid	Octanoic acid	$CH_3(CH_2)_6COOH$
10	Capric acid	Decanoic acid	$CH_3(CH_2)_8COOH$
12	Lauric acid	Dodecanoic acid	$CH_3(CH_2)_{10}COOH$
14	Myristic acid	Tetradecanoic acid	$CH_3(CH_2)_{12}COOH$
16	Palmitic acid	Hexadecanoic acid	$CH_3(CH_2)_{14}COOH$
18	Stearic acid	Octadecanoic acid	$CH_3(CH_2)_{16}COOH$
20	Arachidic acid	Eicosanoic acid	$CH_3(CH_2)_{18}COOH$
22	Behenic acid	Docosanoic acid	$CH_3(CH_2)_{20}COOH$

UNSATURATED FATTY ACIDS

- Monounsaturated
- Polyunsaturated
- Eicosanoids
  - These compounds derived from 20 carbon Fatty acid,
  - Example: Prostaglandins
    - The prostaglandins (PG) are a group of physiologically active lipid compounds having diverse hormone-like effects in animals.

TYPES OF FATTY ACIDS
(according to the number of double bonds)



Saturated (No bond)

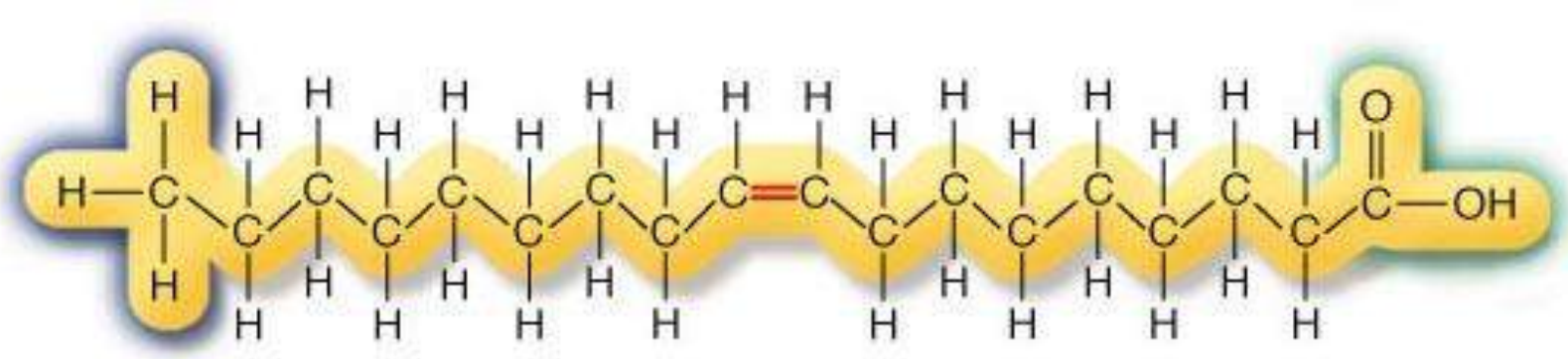


Monounsaturated (1 bond)



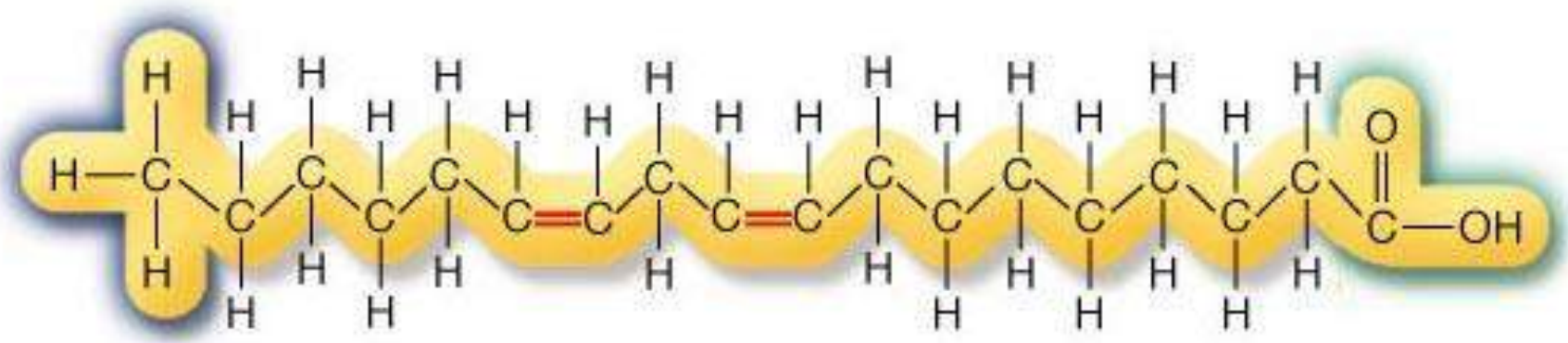
Polyunsaturated (>1 bond)

Monounsaturated Fatty Acid



One carbon-carbon double bond

Polyunsaturated Fatty Acid

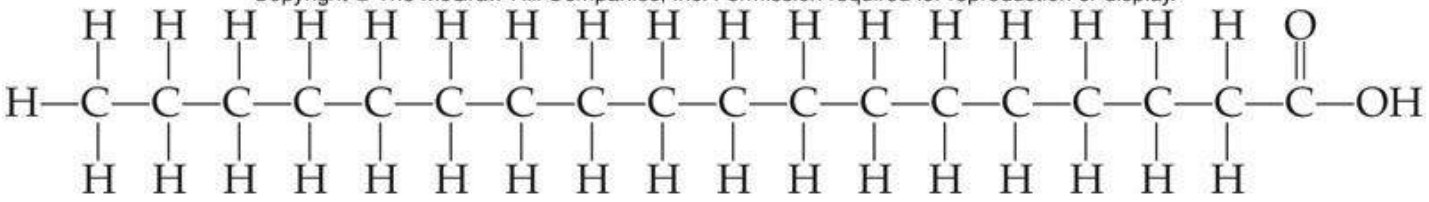


More than one carbon-carbon double bond

Structure

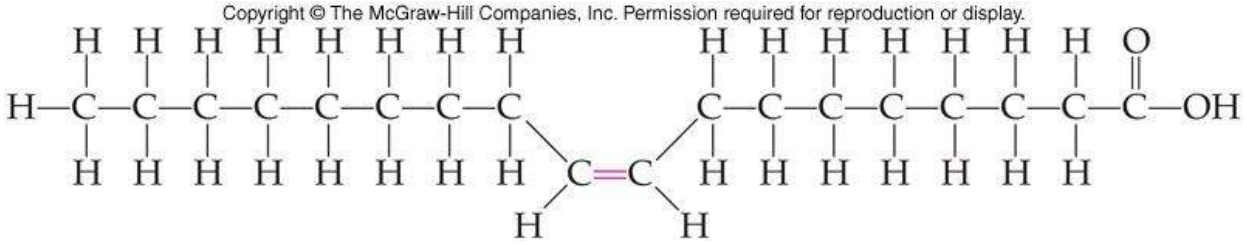
- ***Stearic acid***: a typical saturated fatty acid with 18 carbons in the chain

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- ***Oleic acid***: a typical unsaturated fatty acid with 18 carbons in the chain

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At room temperature, unsaturated fats (such as those found in oil), are usually liquid, whereas saturated fats (such as those found in butter) are solid.

Cis and Trans-Isomers in unsaturated fatty acids

- Depending upon the orientation of the H atoms around the axis of the double bond-
- Cis- If the H atoms are on the same side of the double bond
  - U-like formation; naturally occurring.
- Trans- If the H atoms are on the opposite side
  - more linear; occur in partially hydrogenated foods
- Oleic acid and Elaidic acid have the same formula but Oleic acid is cis while Elaidic acid is Trans Fatty Trans-fatty acid:

FATTY ACIDS: SHAPES

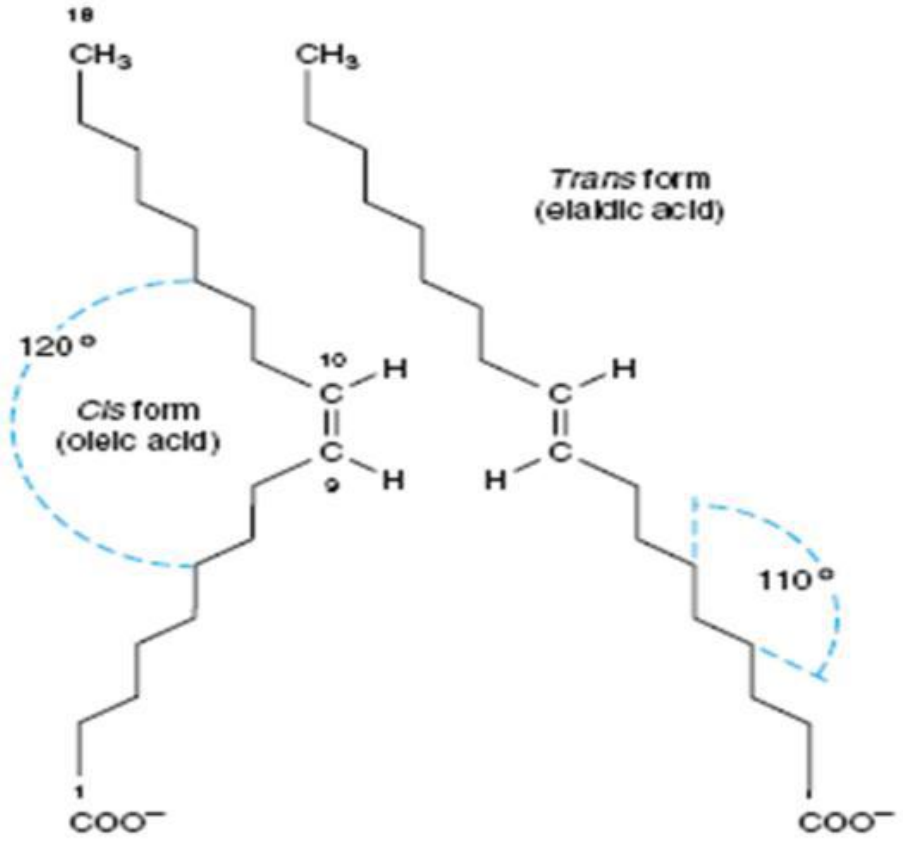
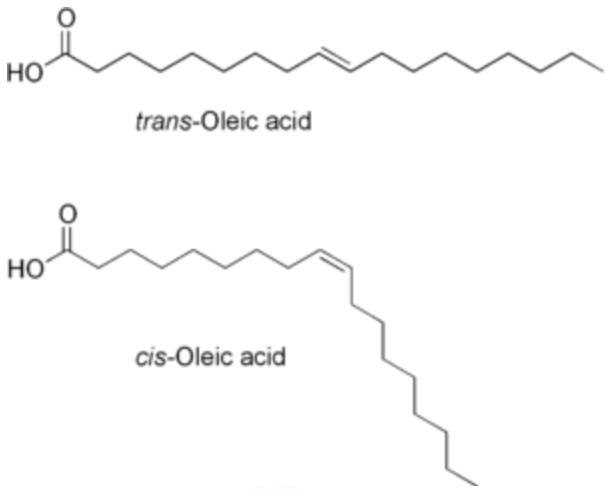
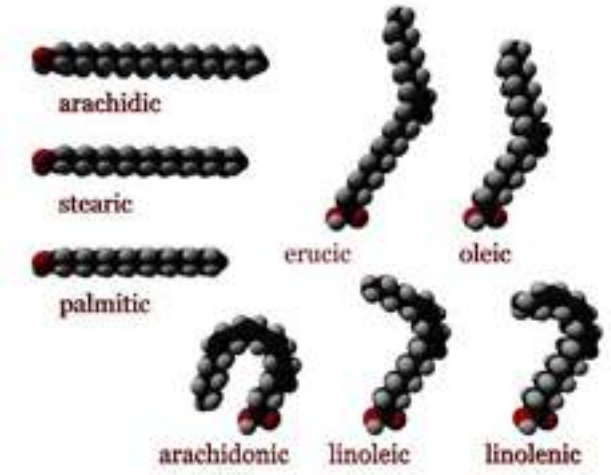


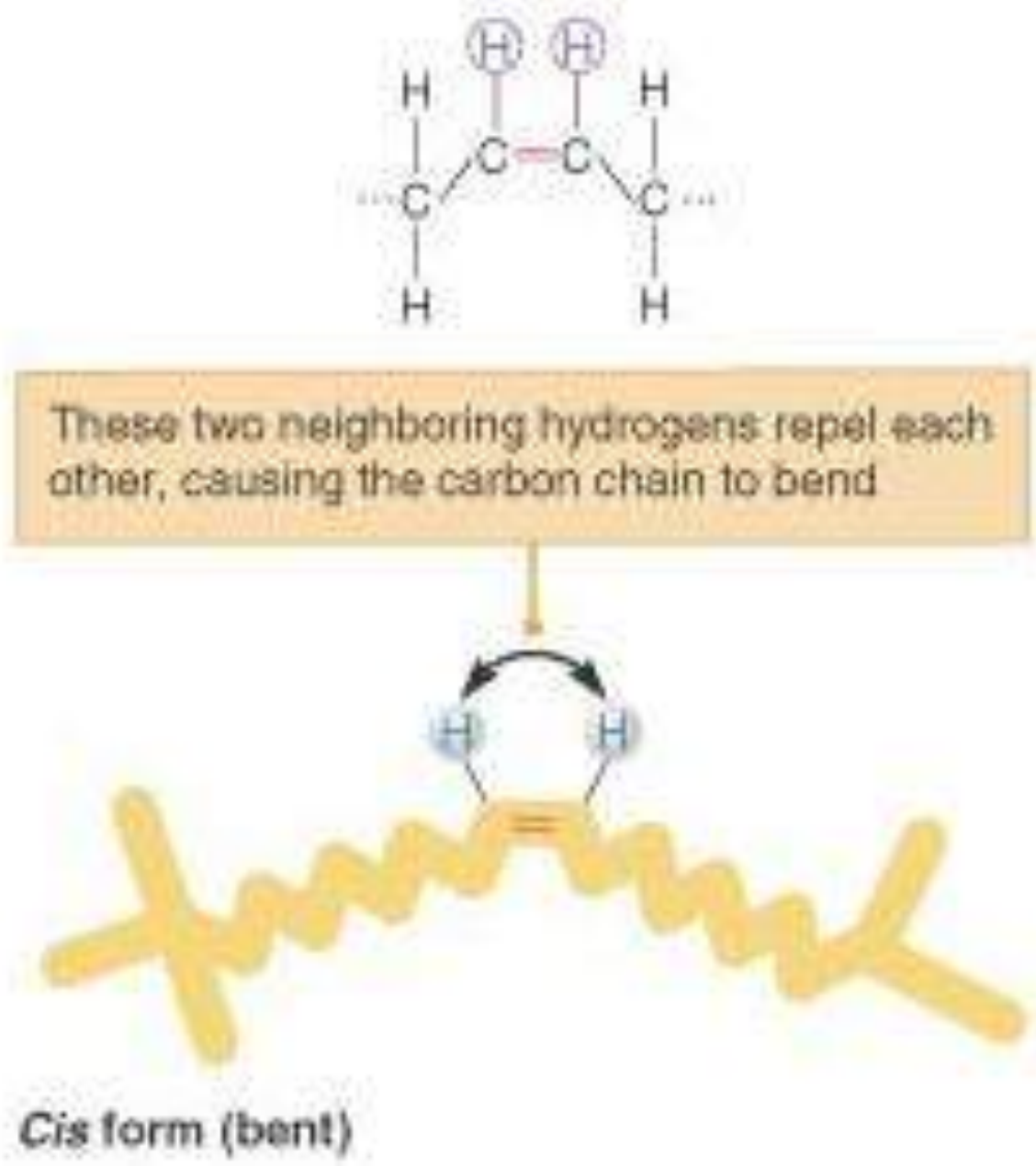
Figure 14–5. Geometric isomerism of Δ^9, 18:1 fatty acids (oleic and elaidic acids).

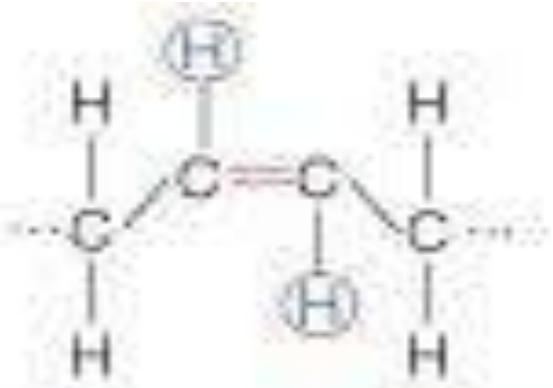
Cis & Trans-Isomers in unsaturated fatty acids

- The hydrocarbon chains in saturated fatty acids are, fairly straight & can pack closely together, making these fats solid at room temperature.
- Oils, mostly from plant sources, have some double bonds between some of the carbons in the hydrocarbon tail, causing bends or **“kinks”** in the shape of the molecules.
- Increase in the number of *cis* double bonds in a fatty acid leads to a variety of possible spatial configurations of the molecule
  - e.g., Arachidonic acid, with four *cis* double bonds, has "kinks" or a U shape.

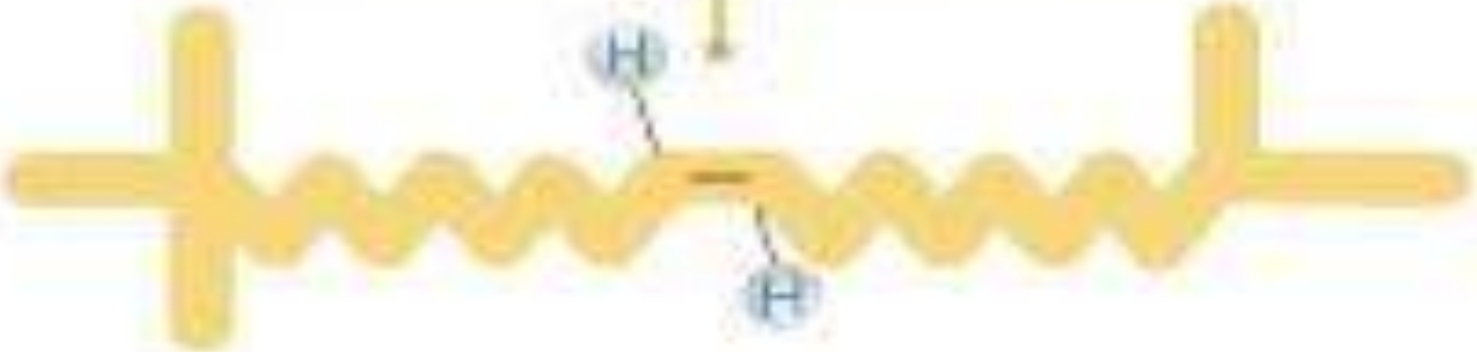





These two neighboring hydrogens repel each other, causing the carbon chain to bend
Cis form (bent)



These two hydrogens are already as far apart as they can get



***Trans* form (straighter)**

Trans Fatty acids

- *Trans* fatty acids are present in certain foods, arising as a by-product of the saturation of fatty acids during hydrogenation, or "hardening," of natural oils in the manufacture of margarine.
- Naturally-occurring unsaturated vegetable oils have almost all cis bonds, but using oil for frying causes some of the cis bonds to convert to trans bonds.

Essential & Non Essential Fatty Acids

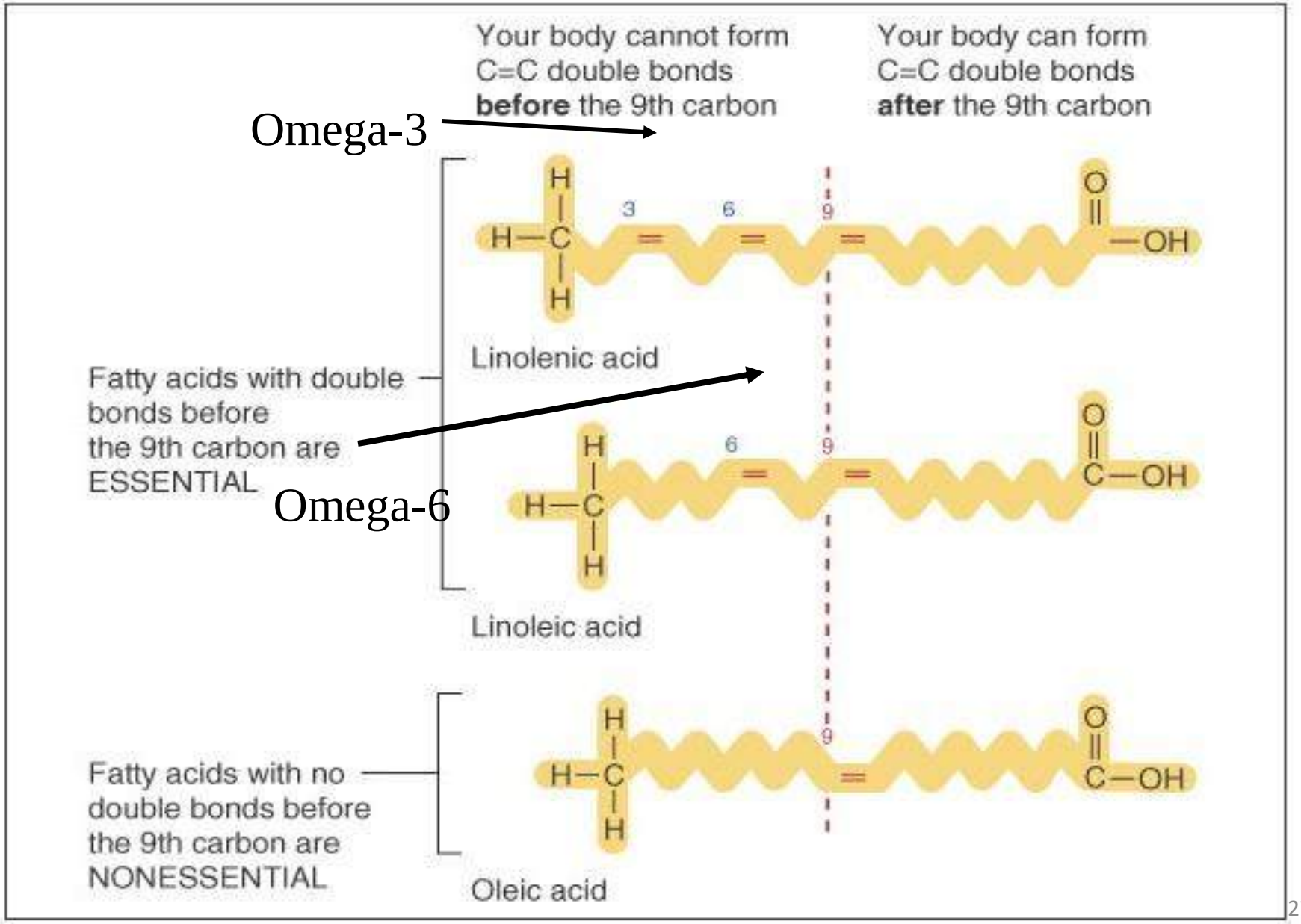
- Essential Fatty Acids
  - our bodies cannot make C-C double bonds before the 9th carbon from the methyl end, so we must get these fatty acids from our diet
  - E.g. Omega-6 linoleic acid & omega-3 alpha-linolenic acid
- Non-Essential Fatty Acids
  - our body can make certain fatty acids so they are not required in the diet

Essential Fatty Acids

- Polyunsaturated fatty acids such as **Linoleic** & **Linolenic acids** are essential for normal life functions. They are therefore characterized as essential fatty acids.
- **Arachidonic acid** is considered as semi essential fatty acid since it can be synthesized from Linoleic acid .
- Essential polyunsaturated fatty acids can be classified as belonging to one of two "families",
  - the **omega-6 family** or the **omega-3 family**.
- Fatty acids belonging to these two families differ not only in their chemistry, but also in their natural occurrence and biological functions.

Location of Double Bonds

- Polyunsaturated fatty acid (PUFA) are identified by position of the double bond nearest the methyl end (CH_3) of the carbon chain; this is described as a omega number.
  - If PUFA has first double bond 3 carbons away from the methyl end => **omega 3 FA**
  - 6 carbons from methyl end => **omega 6 FA**


Your body cannot form C=C double bonds **before** the 9th carbon
Your body can form C=C double bonds **after** the 9th carbon
Omega-3
Fatty acids with double bonds before the 9th carbon are ESSENTIAL
Omega-6
Linolenic acid
Linoleic acid
Fatty acids with no double bonds before the 9th carbon are NONESSENTIAL
Oleic acid

Nomenclature of Fatty acids

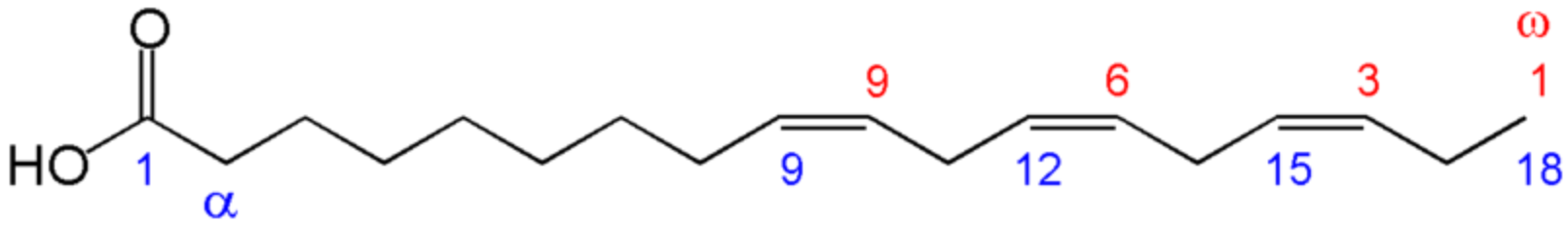
- The systematic name for a fatty acid is derived from the name of its parent hydrocarbon by the substitution of ***oic*** for the final ***e***.
- For example:
  - the C18 saturated fatty acid is called **octadecanoic acid** because the parent hydrocarbon is **octadecane**.
  - A C18 fatty acid
    - with one double bond is called **octadecenoic acid**;
    - with two double bonds, **octadecadienoic acid**;
    - and with three double bonds, **octadecatrienoic acid**.
  - The notation **18:0** denotes a C18 fatty acid with **no** double bonds, whereas **18:2** signifies that there are **two** double bonds.

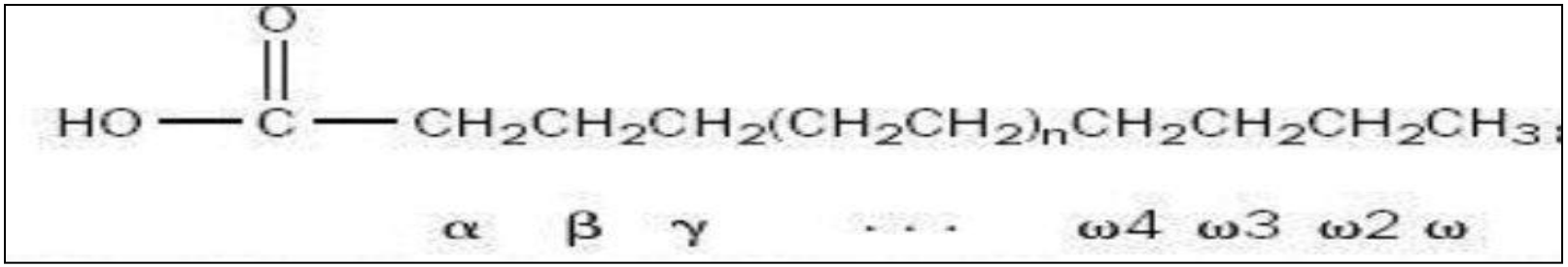
Nomenclature of Fatty acids(Contd.)

- Carbon atoms are numbered from the carboxyl carbon (carbon No. 1).
- The carbon atoms adjacent to the carboxyl carbon No: 2, 3 & 4 are also known as the α ,β & γ carbons, respectively, and the terminal methyl carbon is known as the ω or n-carbon.
- The position of a double bond is represented by the symbol Δ followed by a superscript number.
  - eg, Δ 9 indicates a double bond between carbons 9 and 10 of the fatty acid;

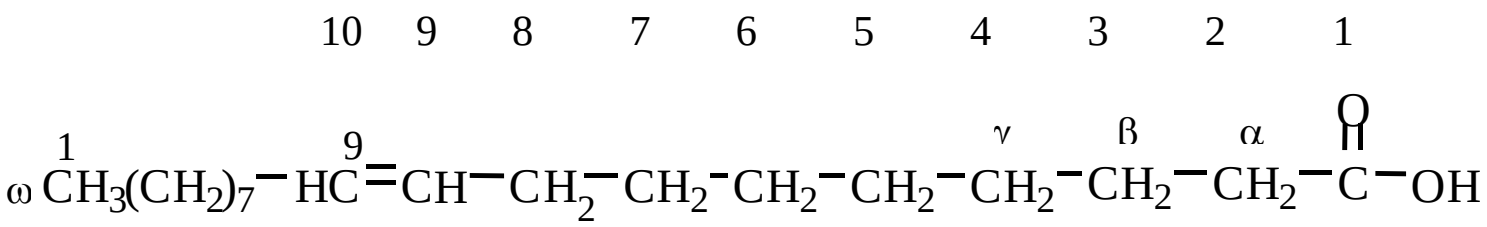
NOMENCLATURE

- Carbon next to carboxylic carbon is alpha (α)
  - Terminal methyl carbon is termed omega (ω)





Naming for Unsaturated FAs



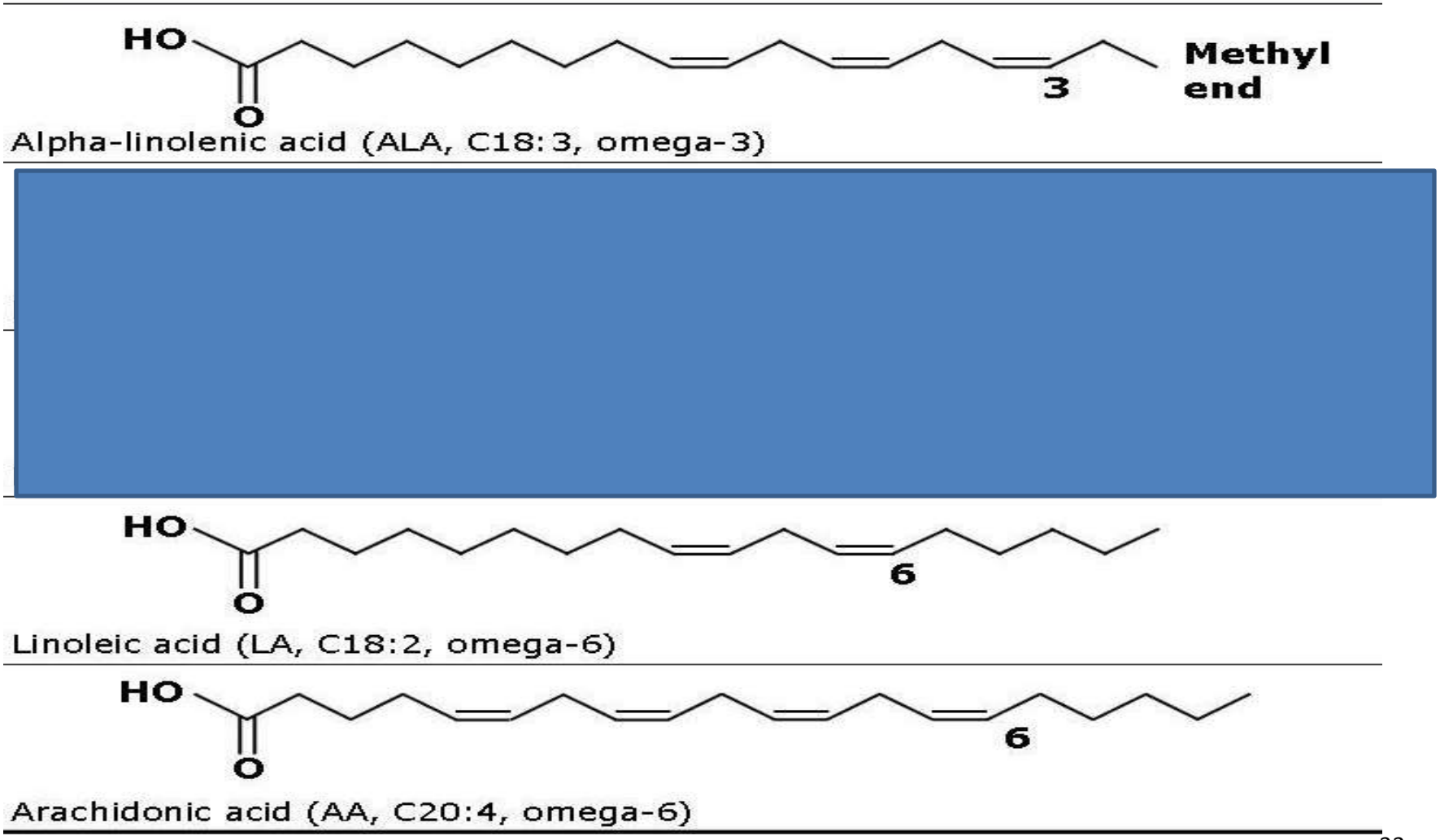
Δ 9, 10 - Octadecenoic acid

9 - Octadecenoic acid

Nomenclature of Fatty acids(Contd.)

- Alternatively, the position of a double bond can be denoted by counting from the distal end, with the ω-carbon atom (the methyl carbon) as number 1.
  - ω9 indicates a double bond on the ninth carbon counting from the ω-carbon.

Omega 3 and Omega 6 fatty acids



Unsaturated Fatty Acids

S.No.	Number of C atoms, number and location of double bonds	Common Name	Systemic Name
[A]	**Monoenoic acids (one double bond)**		
1.	16:1;9	Palmitoleic acid	*cis*-9-Hexadecenoic
2.	18:1;9	Oleic Acid	*cis*-9-Octadecenoic
3.	18:1;9	Elaidic acid	*trans* 9-Octadecanoic
[B]	**Dienoic acids (two double bonds)**		
1.	18:2;9,12	Linoleic acid	all-*cis*-9,12-Octadecadienoic

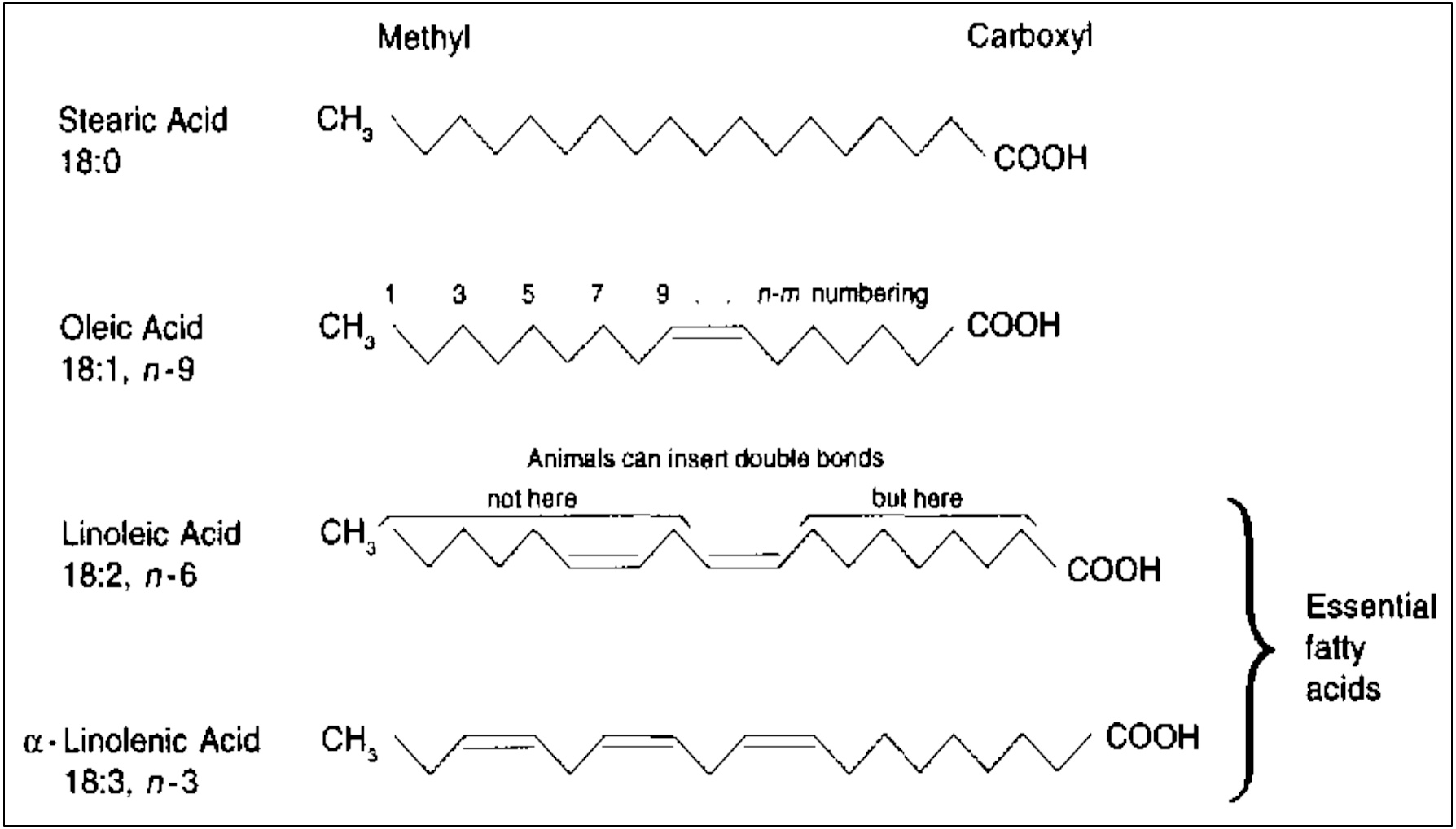
Unsaturated Fatty Acids(Contd.)

S.No.	Number of C atoms, number and location of double bonds	Common Name	Systemic Name
[c]	**Trienoic acids (three double bonds)**		
1.	18:3;6,9,12	Y- Linolenic acid	all-*cis*-6,9,12-Octadecatrienoic
2.	18:3;9,12,15	α-Linolenic	all-*cis*-9,12,15Octadecatrienoic
[D]	**Tetraenoic acid(Four double bonds)**		
	20:4;5,8,11, 14	Arachidonic	all-*cis*-5,8,11,14-Eicosatetraenoic

Unsaturated Fatty Acids(Contd.)

S.No.	Number of C atoms, number and location of double bonds	Common Name	Systemic Name
[E]	**Pentaenoic acids (Five double bonds)**		
1.	20:5;5,8,11,14,17	Timnodonic acid	all-*cis*-5,8,11,14,17-Eicosapentaenoic
[F]	**Hexaenoic acid(six double bonds)**		
	22:6;4,7,10,13,16,19	Cervonic acid	all-*cis*-4,7,10,13,16,19-Docosahexaenoic

Fatty acids


Methyl
Carboxyl
Stearic Acid
18:0
CH3
COOH
1 3 5 7 9 . . . n-m numbering
Oleic Acid
18:1, n-9
CH3
COOH
Animals can insert double bonds
not here
but here
Linoleic Acid
18:2, n-6
CH3
COOH
α-Linolenic Acid
18:3, n-3
CH3
COOH
Essential fatty acids

Biological Importance of fatty acids

- Fatty acids are the building blocks of phospholipids and glycolipids (constituents of cell membranes).
- Fatty acids are attached to many proteins.
- Fatty acids are stored as triacylglycerols (esters of glycerol and fatty acids). Triacylglycerols are also called triglycerides or neutral fats.
- Messenger molecules (messengers). Products of fatty acids function as hormones and as intracellular messenger molecules (messengers).

Some functions of fatty acids

Molecule	Fatty acid	Function
Triacylglycerol	Saturated, unsaturated	Storage of energy Body contours
Local hormones	Eicosanoids., Prostaglandins	Inflammatory response Inflammation is a protective response that involves immune cells, blood vessels, and molecular mediators. The purpose of inflammation is to eliminate the initial cause of cell injury, clear out necrotic cells and tissues damaged from the original insult and the inflammatory process, and to initiate tissue repair.
Phospholipids	Saturated, unsaturated FA	Membranes (cell, organelles
Sphingolipids	Saturated FA	Myelination the process of forming a myelin sheath around a nerve to allow nerve impulses to move more quickly.

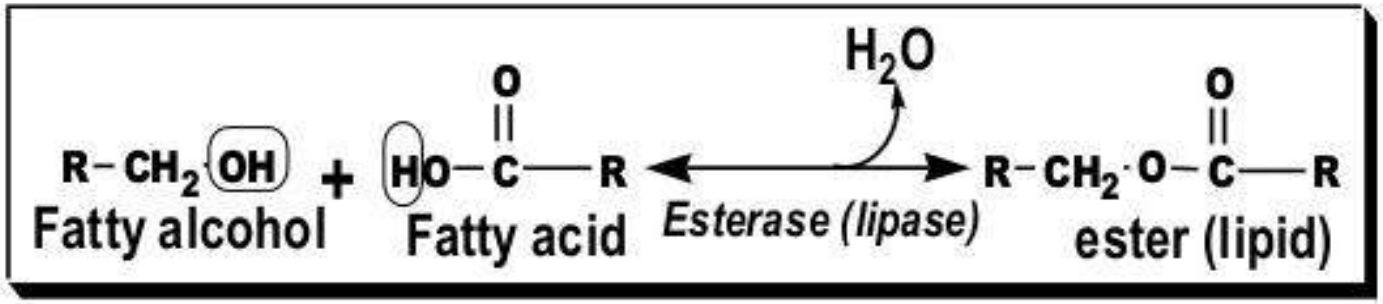
Lipids

- The lipids are a heterogeneous group of compounds, including fats, oils, steroids, waxes, and related compounds, that are related more by their physical than by their chemical properties.
- They have the common property of being
  - (1) relatively **insoluble in water** and
  - (2) **soluble in nonpolar solvents** such as ether and chloroform.

Organic substances relatively insoluble in water but soluble in organic solvents like chloroform, ether and benzene

Chemistry of Lipids

- Definition
  - Lipids are organic compounds
  - formed mainly from alcohol and fatty acids combined together by ester linkage.



FATTY ACIDS

- Mono carboxylic acids with a hydrocarbon chain of varying length



- Hydrocarbon chain with two ends
  - Carboxylic acid
  - Methyl group

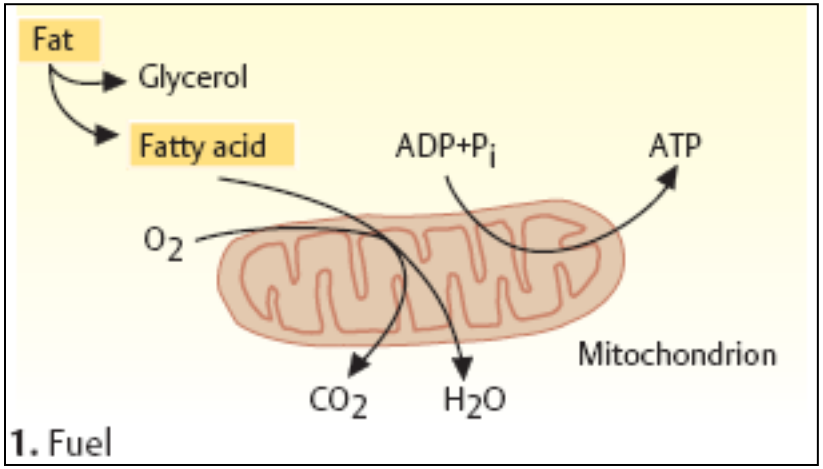
Functions of Lipids

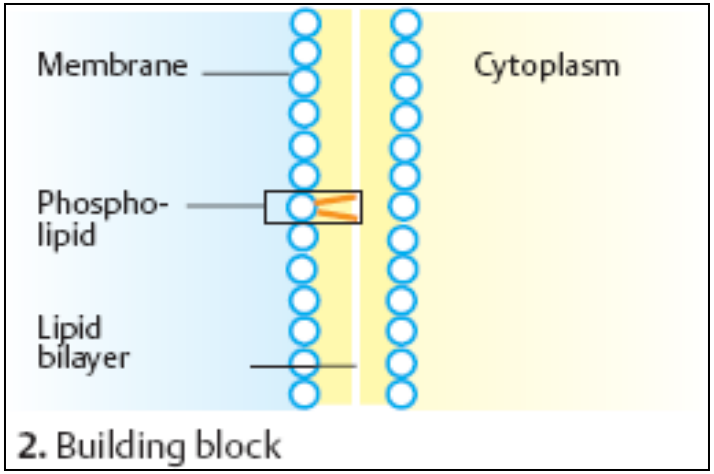
- Storage form of energy
- Important dietary components
  - because of their high energy value and also because of the fat-soluble vitamins and the essential fatty acids contained in the fat of natural foods.
- Structural components of cell membranes
- Serve as thermal insulators in the subcutaneous tissues and around certain organs

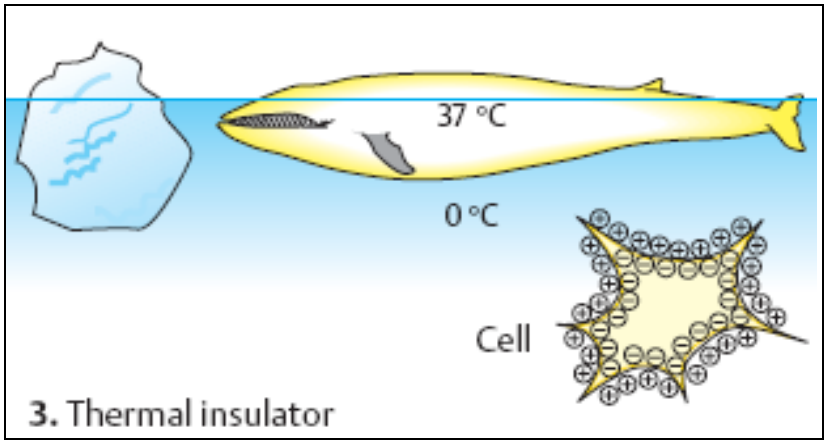
Functions of Lipids

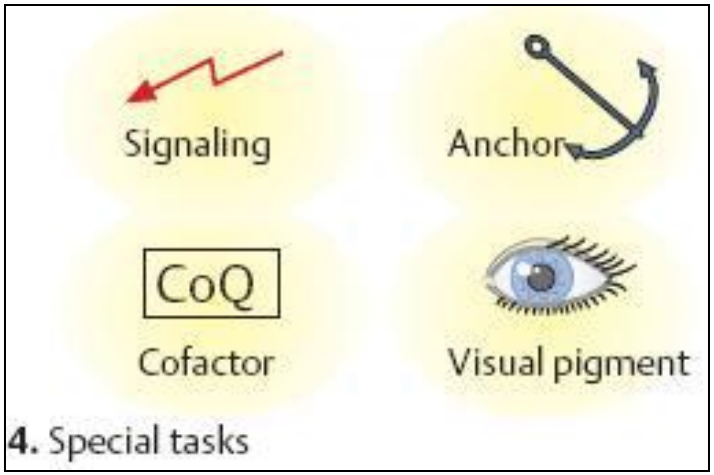
- Nonpolar lipids act as electrical insulators
  - allowing rapid propagation of depolarization waves along myelinated nerves Provide shape and contour to the body
- Lipoproteins are important cellular constituents, serving as means of transporting lipids.

FUNCTIONS OF LIPIDS

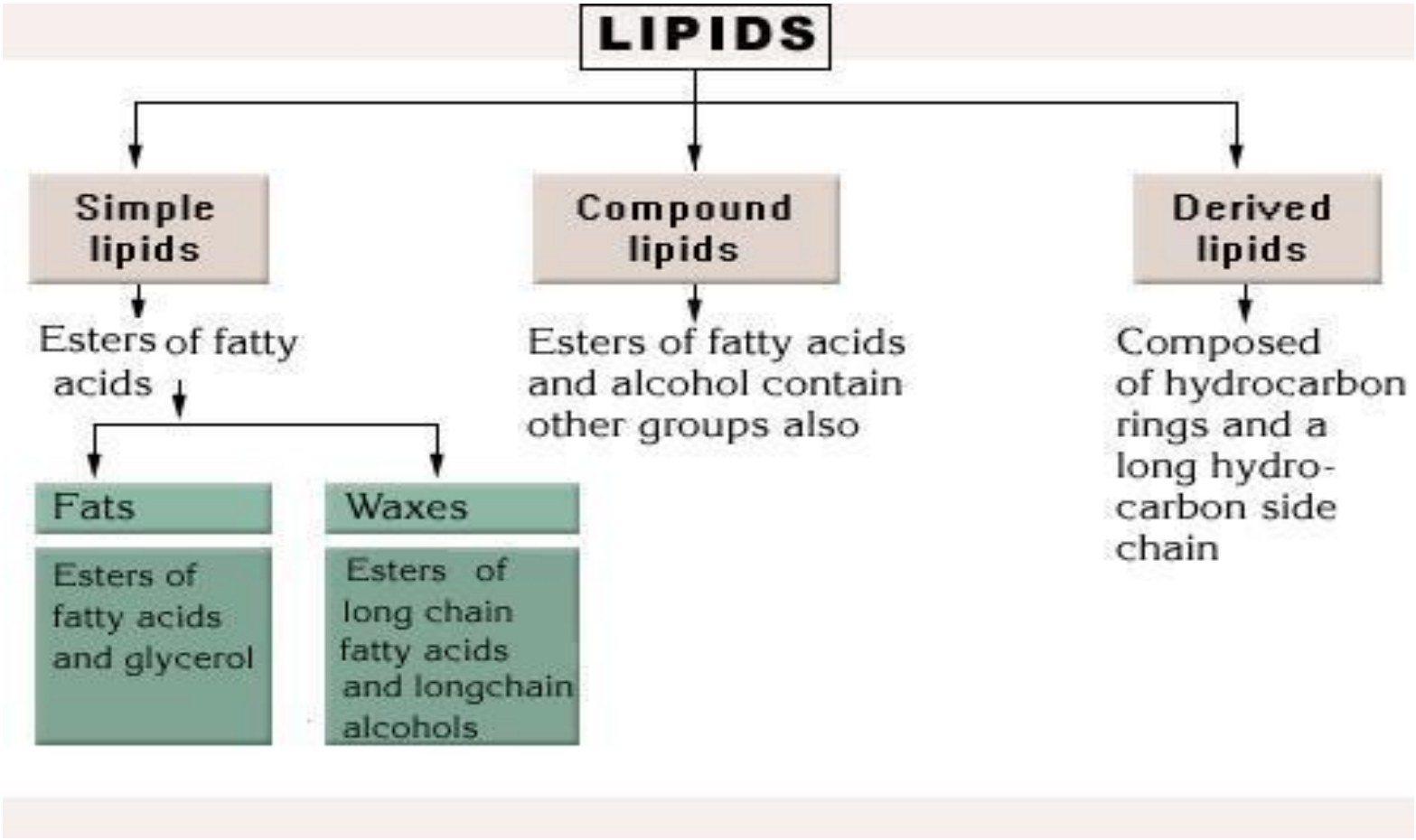








CLASSIFICATION

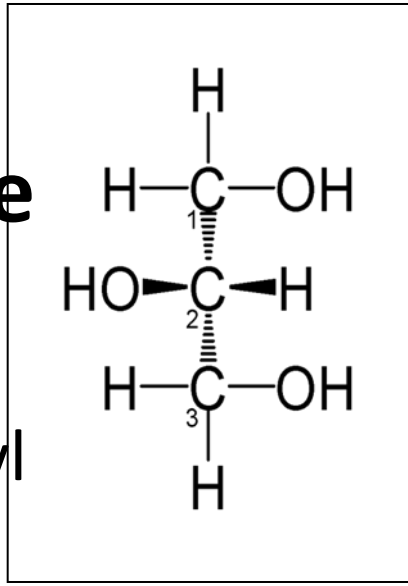


Simple Lipids

- **Simple Lipids**
  - They are esters of fatty acids with various alcohols
  - Depending on the type of alcohol they are classified as

1. Neutral fats or oils (Triacylglycerol)
   - Fats: Esters of fatty acids with glycerol. Oils are fats in the liquid state.
   - Alcohol is GLYCEROL
2. Waxes
   - Esters of fatty acids with higher molecular weight monohydric alcohols.
   - Alcohol is other than glycerol

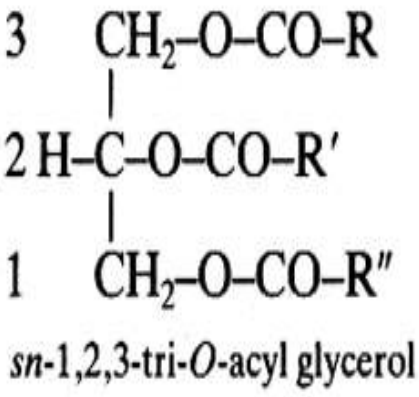
Glycerol-Structure and significance



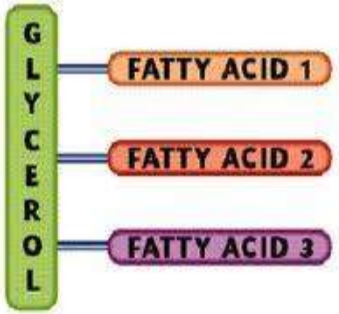
- Also called ‘Glycerin’.
- Trihydric alcohol as it contains three hydroxyl groups
- Can be obtained from diet, from lipolysis of fats in adipose tissue and from glycolysis.
- Can be utilized for the synthesis of triacylglycerols, phospholipids, glucose or can be oxidized to provide energy
- Used as a solvent in the preparation of drugs and cosmetics

Simple Lipids : Neutral fats or Triacylglycerides

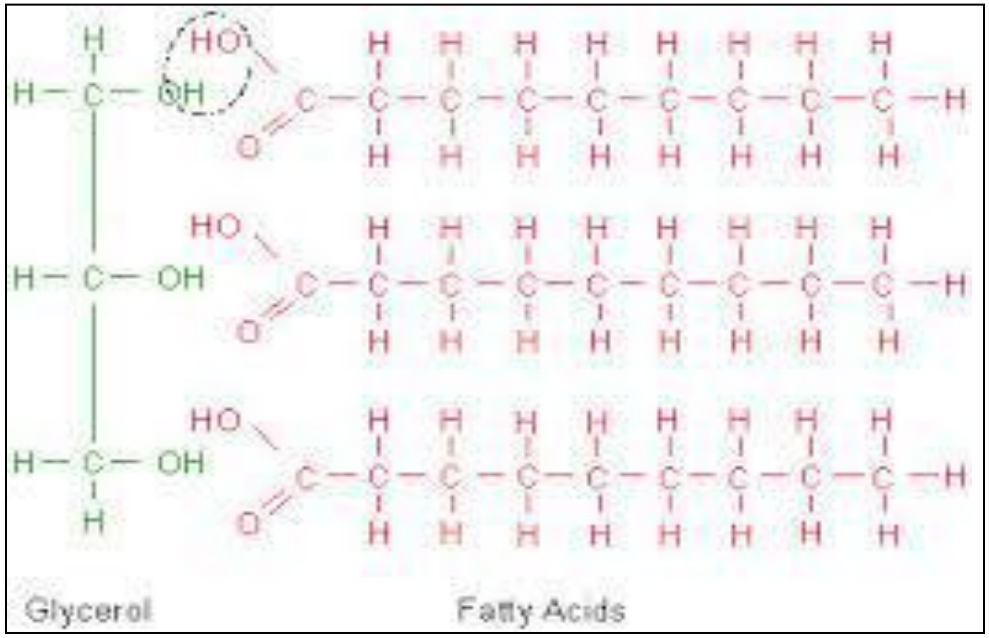
- The triacylglycerols are esters of the trihydric alcohol, glycerol and fatty acids.
- Mono- and Diacylglycerol
  - wherein one or two fatty acids are esterified with glycerol, are also found in the tissues.
- Naturally occurring fats and oils are mixtures of triglycerides.
- If all the OH groups are esterified to same fatty acids- It is **Simple Triglyceride**
- If different fatty acids are esterified- it is known as **Mixed triglyceride.**
- **Polyunsaturated fatty acid is esterified at 2nd position.**

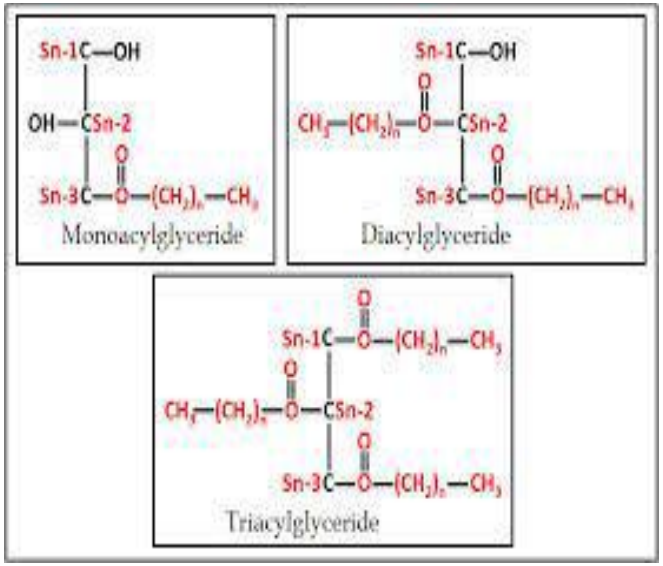


sn-1,2,3-tri-*O*-acyl glycerol



Triacylglycerides





•In a fat, three fatty acids are joined to glycerol by an ester linkage, creating a triacylglycerol, or triglyceride

Properties of triglycerides

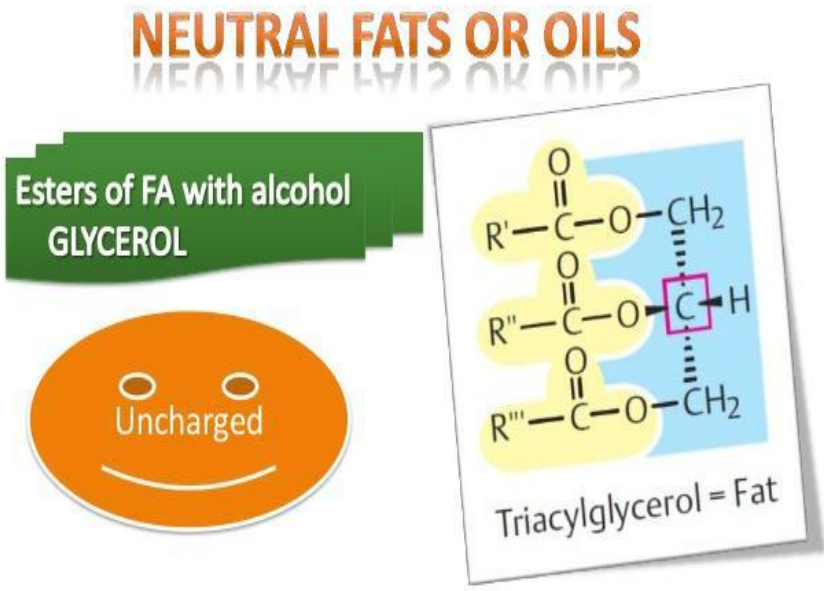
- Colorless, odorless and tasteless
- Insoluble in water
- Oils are liquids at 20^0C, they contain higher proportion of Unsaturated fatty acids
- Fats are solid at room temperature and contain saturated long chain fatty acids
- Triglycerides are the storage form of energy in adipose tissue
- Triglycerides in the body are hydrolyzed by Lipases
  - Lipases are enzymes which catalyze hydrolysis of triglycerides yielding fatty acids & glycerol

Waxes

- They are esters of higher fatty acids with higher MW mono hydroxy aliphatic alcohols(e.g. Cetyl alcohol)
- Have very long straight chain of 60-100 carbon atoms
- In nature, they are found on the surface of plants and insects.

FUNCTIONS OF SIMPLE LIPIDS

- Neutral lipids
- Stored in adipose tissue
- Give body contours
- Insulators
- Store energy

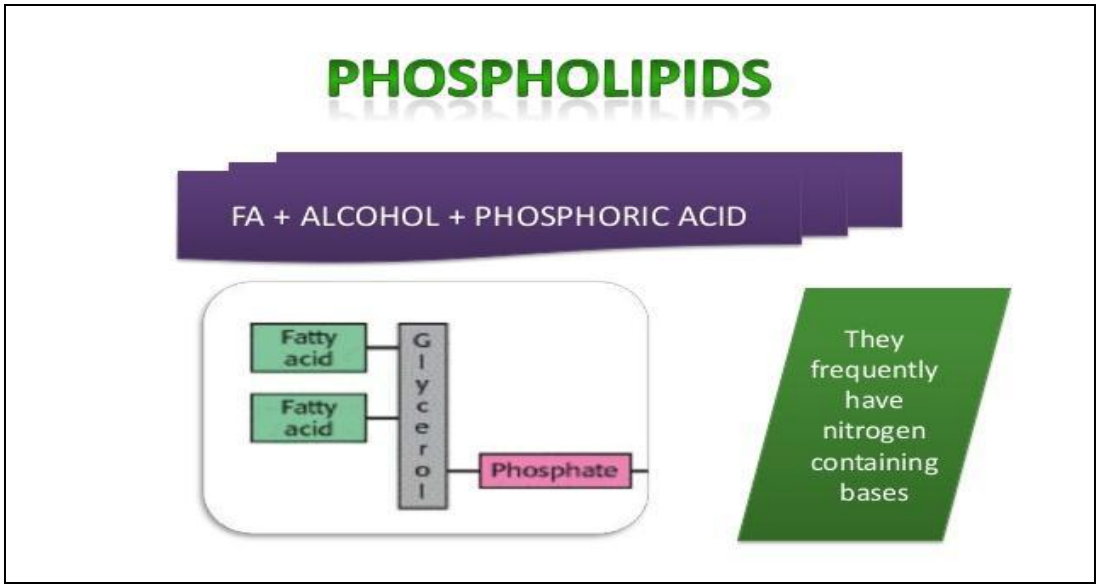


Complex or Compound Lipids

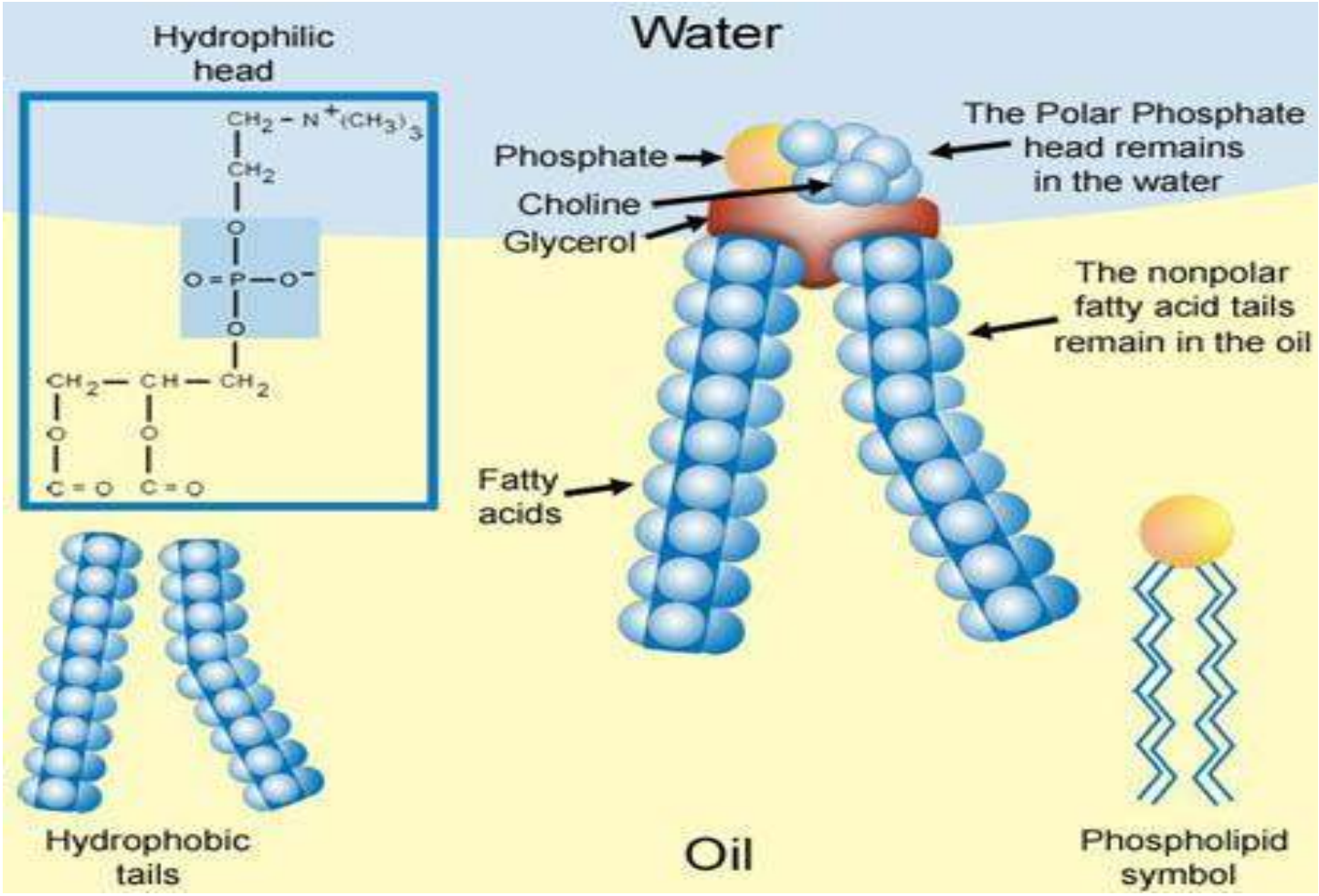
- Complex Lipids
  - They are esters of fatty acids with alcohols containing additional (prosthetic) groups.
- Sub classified according to the type of additional group
  1. Phospholipids
  2. Glycolipids (glycosphingolipids)
  3. Lipoproteins

1. Phospholipids

- Phospholipids
  - Contain in addition to fatty acids and glycerol/or other alcohol, a phosphoric acid residue, nitrogen containing base and other substituents.
  - They are amphipathic molecules containing a polar head and a hydrophobic portion



Phospholipids

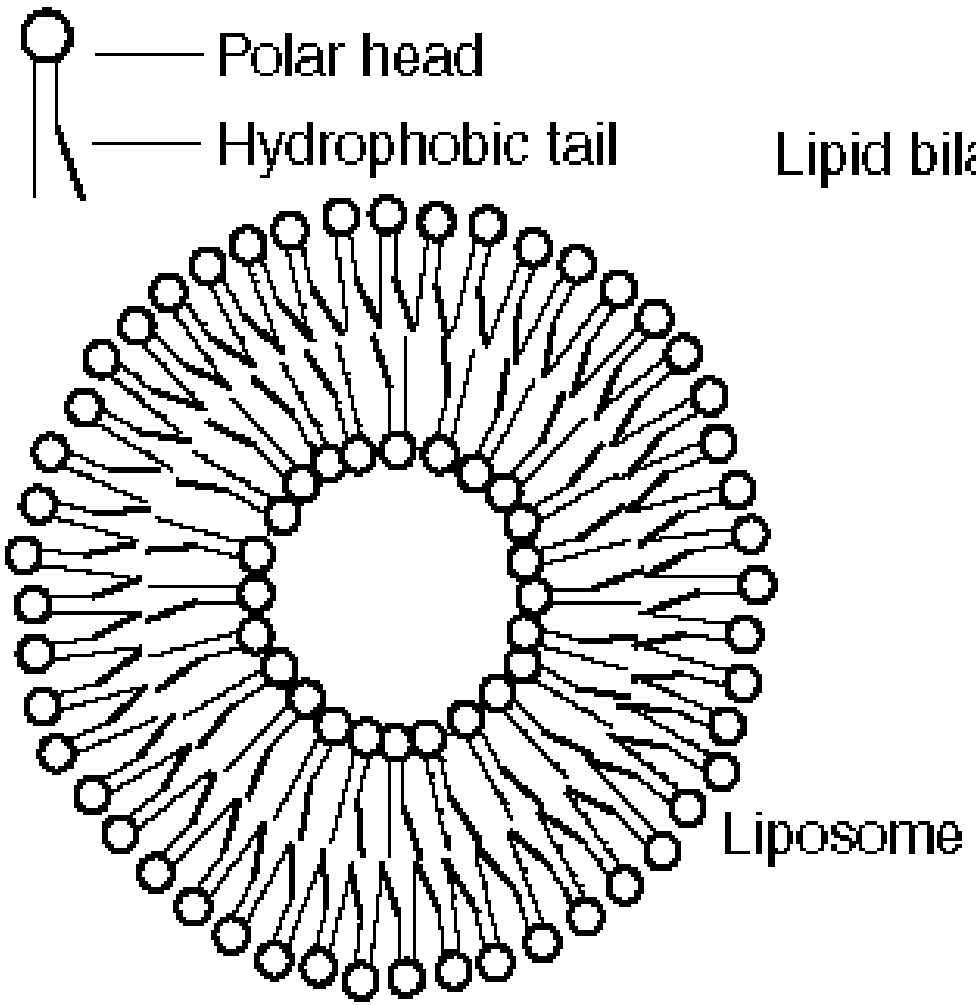


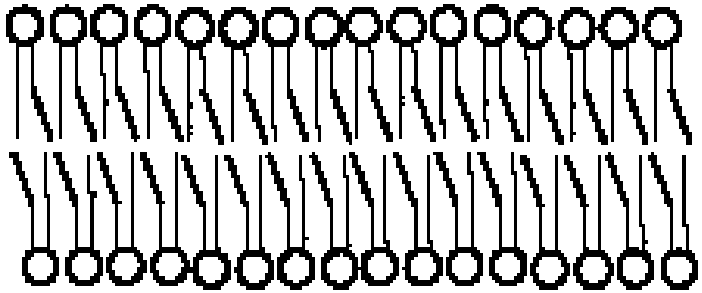
Amphipathic lipids

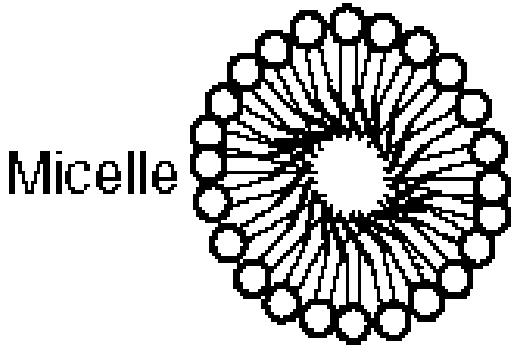
- Fatty acids, phospholipids, sphingolipids, bile salts, and, to a lesser extent, cholesterol contain polar groups. Therefore, part of the molecule is hydrophobic, or water-insoluble; and part is hydrophilic, or water-soluble. Such molecules are described as amphipathic
- They become oriented at oil:water interfaces with the polar group in the water phase and the nonpolar group in the oil phase.
- A bilayer of such amphipathic lipids is the basic structure in biologic membranes

Amphipathic lipids

Interactions of Phospholipids in Aqueous Media







Classification of phospholipids

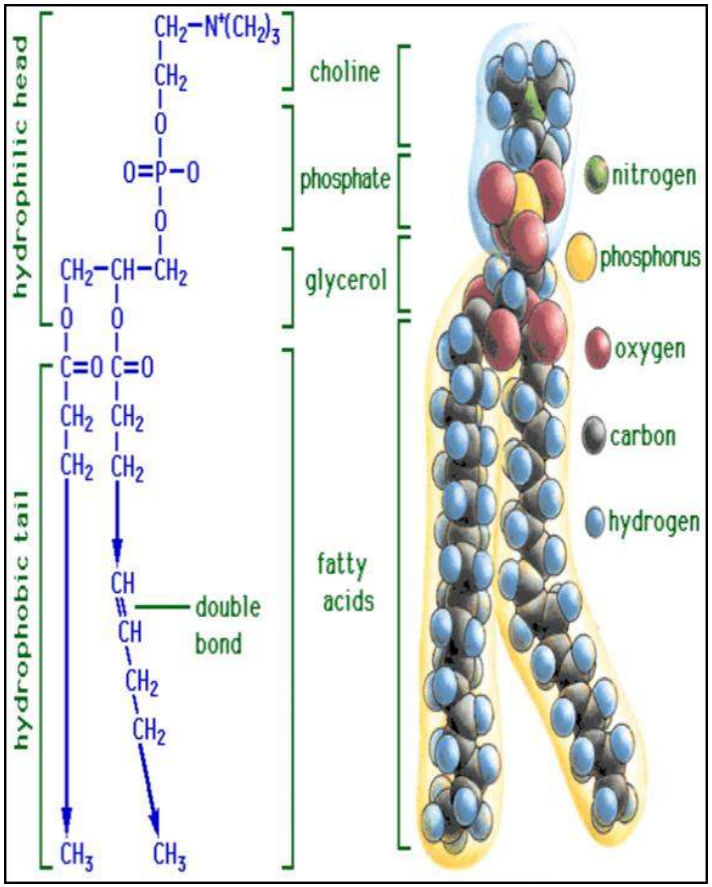
- **Based on nature of alcohol-**

1. **Glycerophospholipids- Glycerol is the alcohol group**

- **Examples-**
  - Phosphatidyl choline
  - Phosphatidyl ethanolamine
  - Phosphatidyl serine
  - Phosphatidyl inositol
  - Phosphatidic acid
  - Cardiolipin
  - Plasmalogen
  - Platelet activating factor
  - Phosphatidyl Glycerol

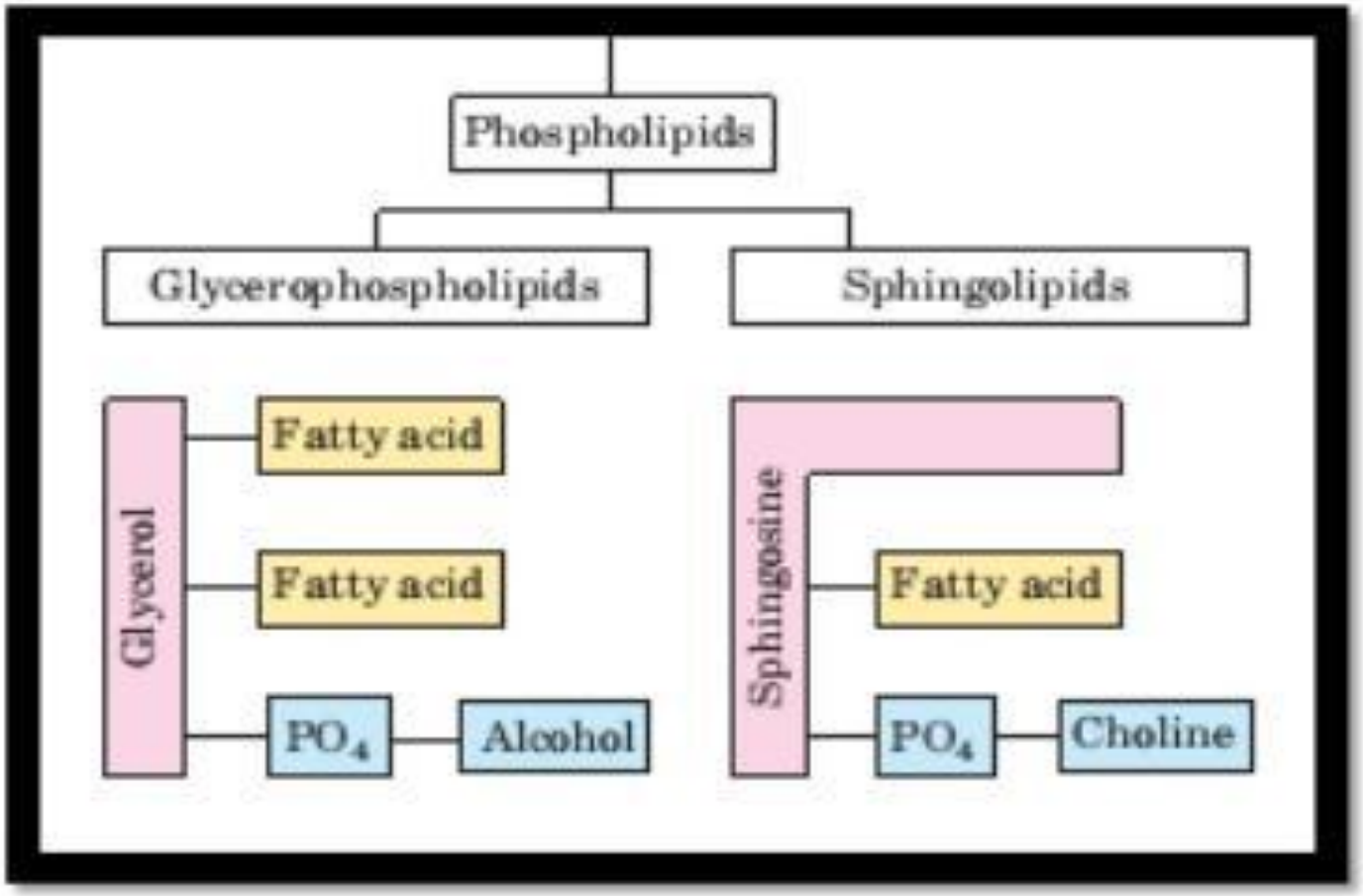
2. Sphingophospholipids- Sphingol is the alcohol group

- Example- Sphingomyelin



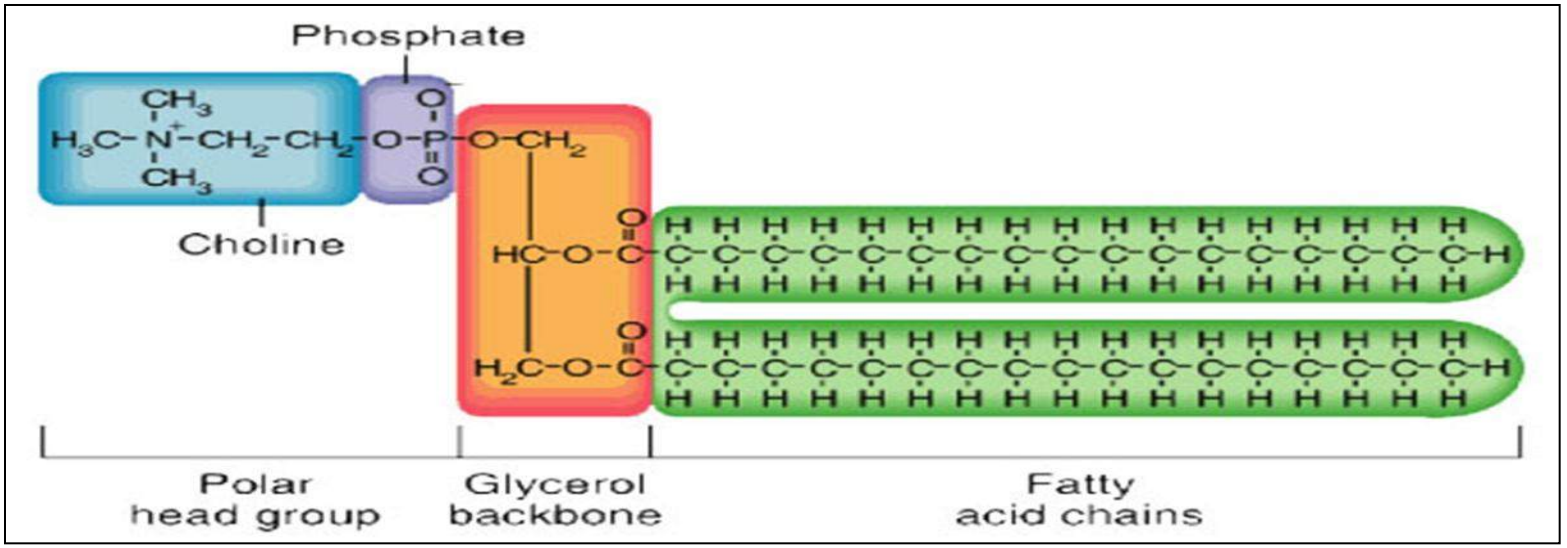
Both classes are found as structural components of membranes

Phospholipid Classification



Glycerophospholipids

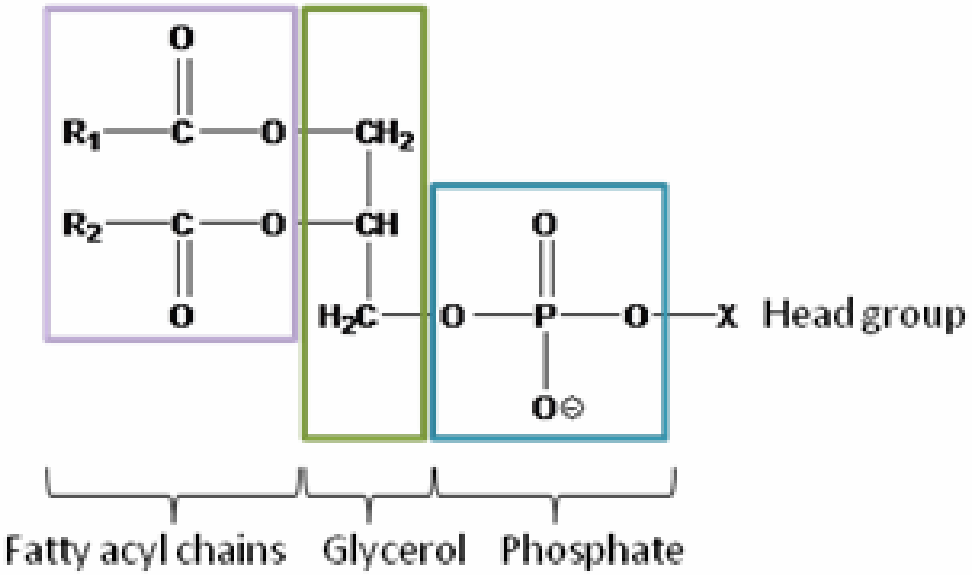
- **Phosphatidylcholines (Lecithins)**
  - Phosphoacylglycerols containing choline are the most abundant phospholipids of the cell membrane
  - Are present a large proportion of the body's store of choline. Choline is important in nervous transmission, as acetylcholine, and as a store of labile methyl groups



Glycerophospholipids

- **Phosphatidyl ethanolamine (cephalin)-**
  - Structurally similar to Lecithin with the exception that the base Ethanolamine replaces choline
  - Brain and nervous tissue are rich in Cephalin
- **Phosphatidyl Serine-**
  - (found in most tissues) differ from phosphatidylcholine only in that serine replaces choline
- **Phosphatidylinositol -Phosphatidylinositol 4,5-bisphosphate**
  - is an important constituent of cell membrane phospholipids; upon stimulation by a suitable hormone agonist, it is cleaved into **diacylglycerol** and **inositol trisphosphate,** both of which act as internal signals or second messengers.
- **Cardiolipin –**
  - Abundantly found in mitochondrial membrane.
- **Plasmalogens –**
  - constitute as much as 10% of the phospholipids of brain and muscle.

Glycerophospholipid- structures



X = —H phosphatidic acid

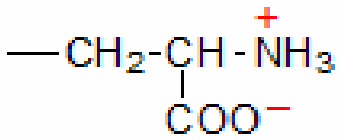


phosphatidylcholine

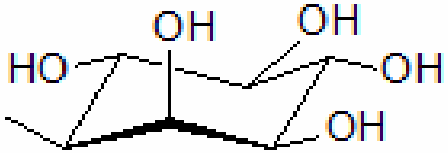


phosphatidylethanolamine

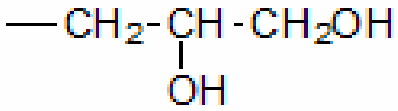
X =



phosphatidylserine



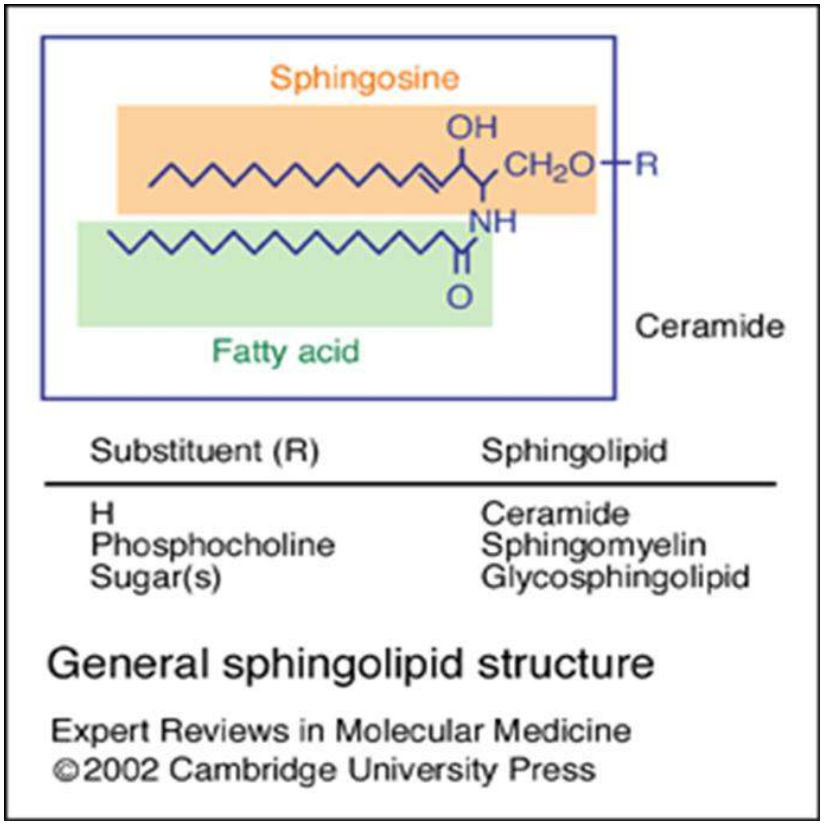
phosphatidylinositol

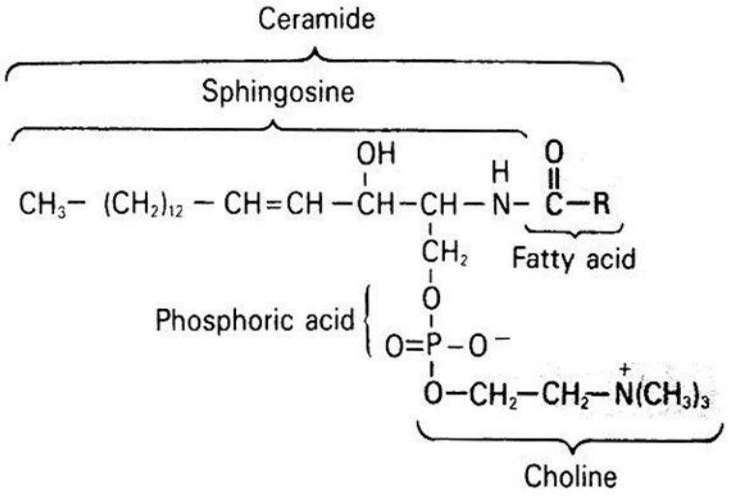


phosphatidylglycerol

Sphingophospholipids

- **Sphingomyelin-**
  - Backbone is sphingosine (amino alcohol)
  - A long chain fatty acid is attached to amino group of sphingosine to form Ceramide
  - The alcohol group of sphingosine is esterified to phosphoryl choline, producing sphingomyelin
  - Sphingomyelin is an important component of myelin of nerve fibers





Functions of Phospholipids

- Components of cell membrane, mitochondrial membrane and lipoproteins
- Participate in lipid absorption and transportation from intestine
- Play important role in blood coagulation
- Required for enzyme action- especially in mitochondrial electron transport chain
- Act as reservoir of second messenger- Phosphatidyl Inositol
- Act as cofactor for the activity of Lipoprotein lipase
- Phospholipids of myelin sheath provide insulation around the nerve fobers

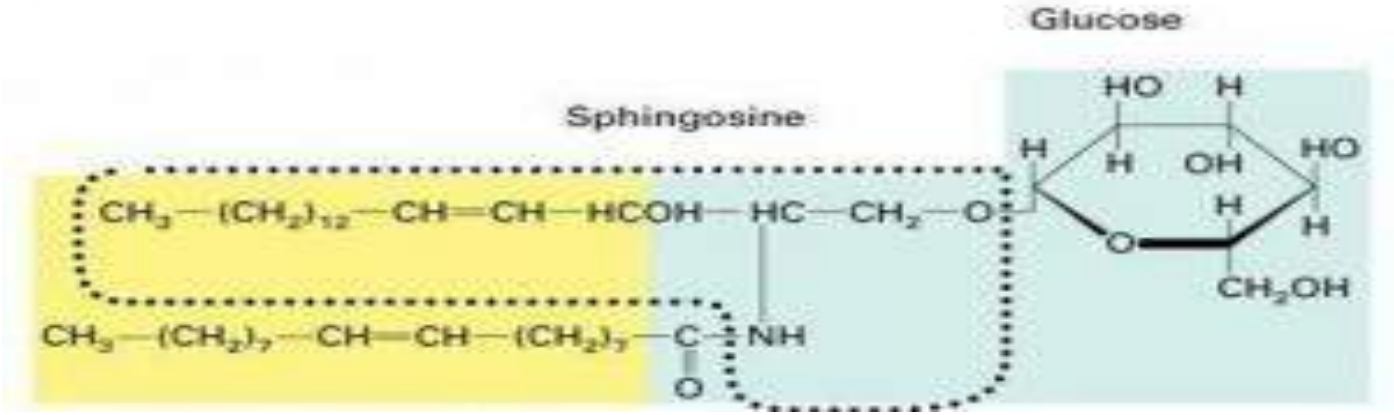
2. Glycolipids

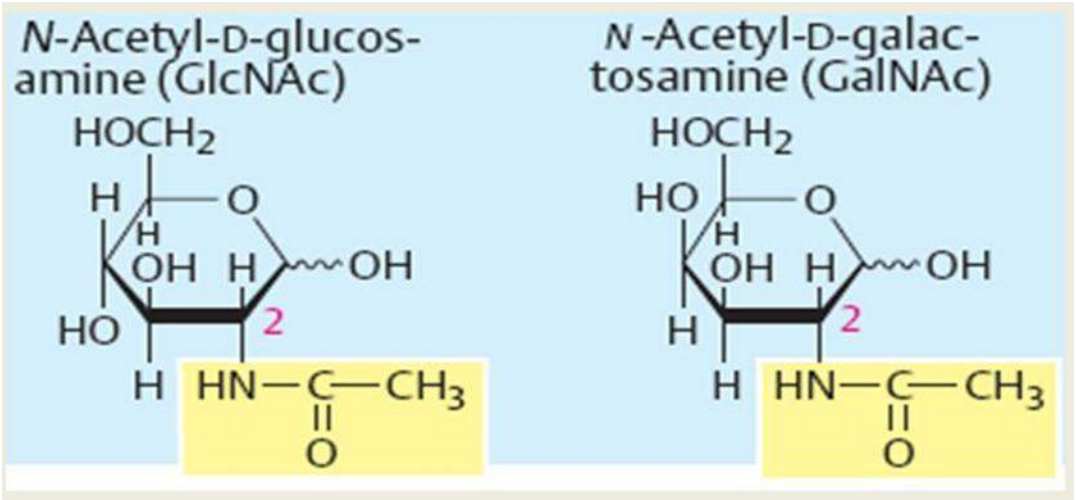
- Glycolipids(glycosphingolipids)
  - Lipid containing a fatty acid, sphingosine and carbohydrate
    - FA + ALCOHOL[SPINGOSINE] +CARBOHYDRATE WITH NITROGEN BASE
- They do not contain phosphate group
- The number and type of carbohydrate moieties present, determine the type of glycosphingolipid.
- There are two types of Glycolipids-
  - A) Neutral glycosphingolipids (Example: Cerebrosides)
  - B) Acidic glycosphingolipids (Example: Gangliosides)

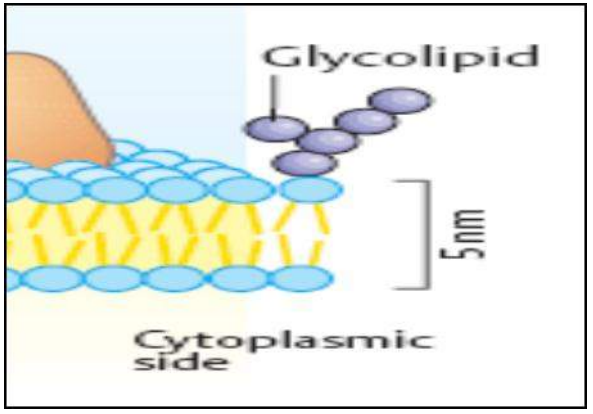
a) Neutral Glycosphingolipids

- **Cerebrosides-** These are ceramide monosaccharides, that contain either a molecule of galactose(Galactocerebroside)or glucose(Glucocerebroside)
- Found predominantly in the brain and nervous tissue with high concentration in myelin sheath
- Ceramide oligosaccharides (Globosides) are produced by attaching additional monosaccharides to Glucocerebroside.
- Lactosyl ceramide contains lactose (Galactose and Glucose attached to ceramide)

Structure- Glucosyl ceramide







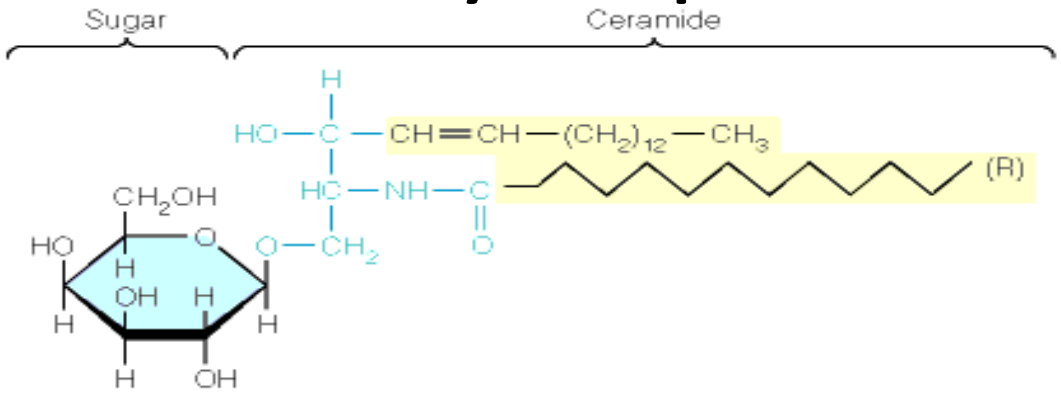
b) Acidic Glycosphingolipids

- They are negatively charged at physiological pH
- The negative charge is imparted by N- acetyl Neuraminic acid(Sialic acid)
- Classified as
  - Ganglioside
  - Sulfatide (Sulfoglycosphingolipids)

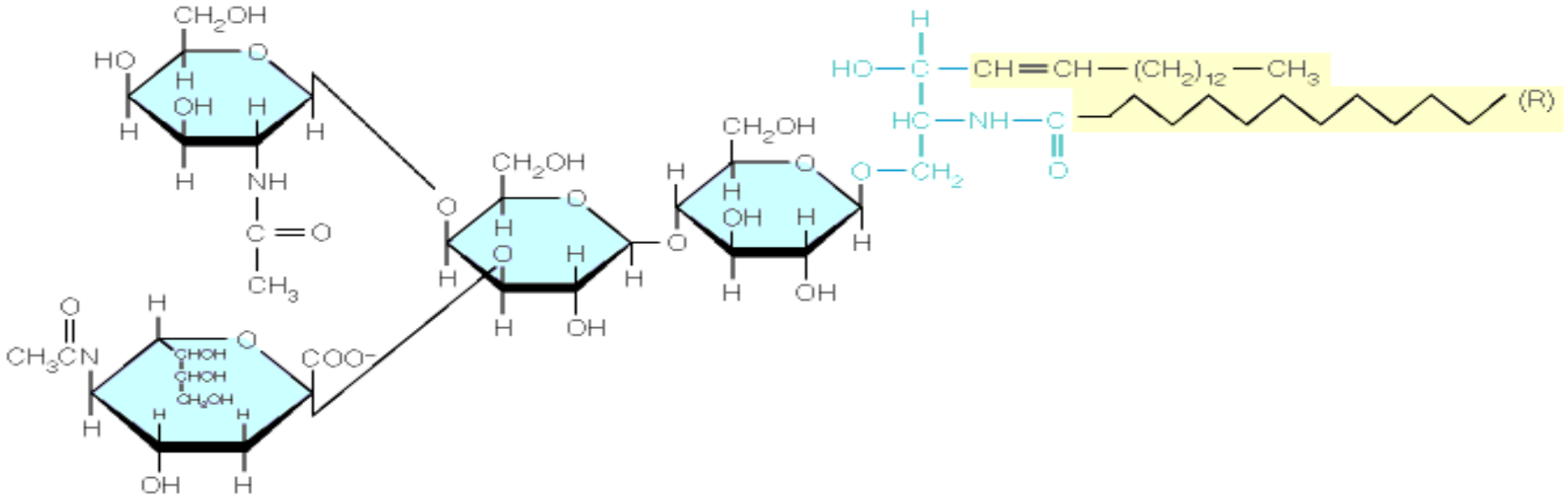
b) Acidic Glycosphingolipids (Gangliosides)

- Brain gangliosides may contain up to four Sialic acid residues and based on that they are-GM, GD, GT and GQ, containing 1,2,3 or 4 Sialic acid residues
- Four important types of Gm series are-
  - GM1, GM2 and GM3
  - GM1 is complex of all

Glycolipids- Structures



(a) Galactosylceramide



(b) GalNAcβ(1→4)Galβ(1→4)Glcβ(1→1)ceramide

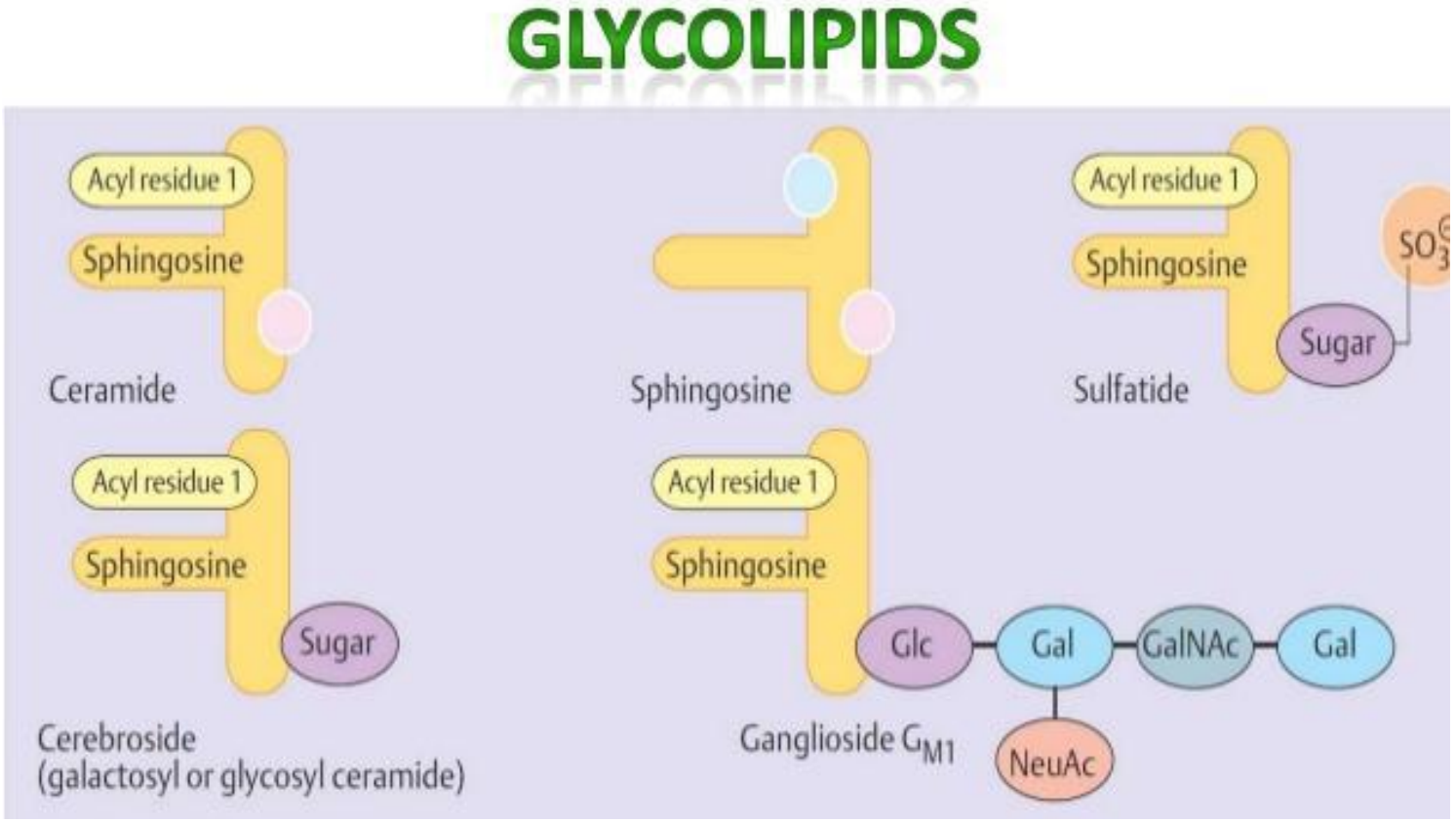
Functions of Glycosphingolipids

- They occur particularly in the outer leaflet of the plasma membrane, where they contribute to **cell surface carbohydrates.**
- They act as cell surface receptors for various hormones, and growth factors
- Play important role in cellular interactions, growth and development
- They are source of blood group antigens and various embryonic antigens

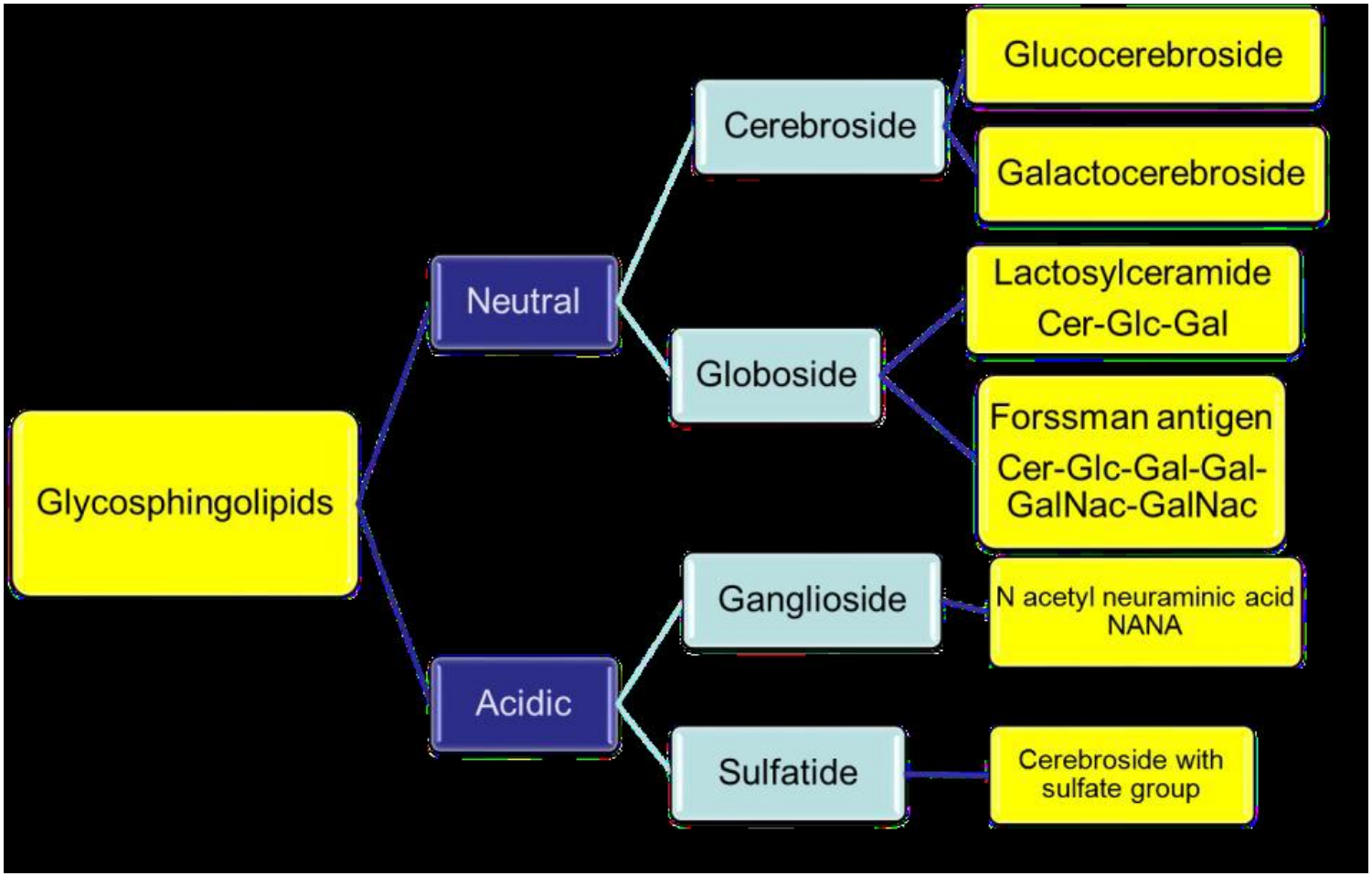
b) Sulfolipids(Sulfoglycosphigolipids)

- They are cerebrosides that contain sulfated galactosyl residues
- Negatively charged at physiological pH
- Found predominantly in nerve tissue and kidney
- Failure of degradation causes them to accumulate in nervous tissues

Glycolipids- Structures



CLASSIFICATION OF GLYCOLIPIDS

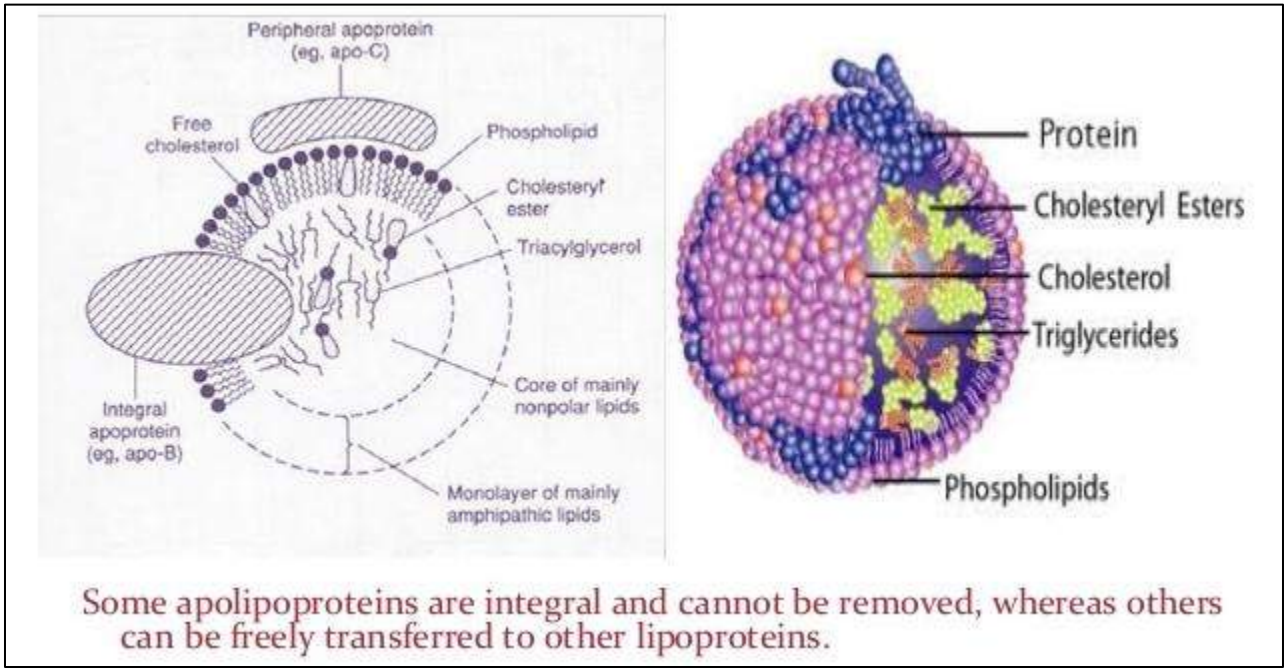


LIPOPROTEINS:

- LIPOPROTEINS:
  - A lipoprotein is a biochemical assembly that contains both proteins and lipids.
  - bound to the proteins, which allow fats to move through the water inside and outside cells.
  - The proteins serve to emulsify the lipid molecules

LIPOPROTEIN STRUCTURE

- Phospholipid membrane
- Membrane proteins called Apolipoproteins
- Core
  - Triacylglycerol (TAG)
  - Cholesterol
  - Cholesteryl Esters



CLASSIFICATION OF LIPOPROTEIN

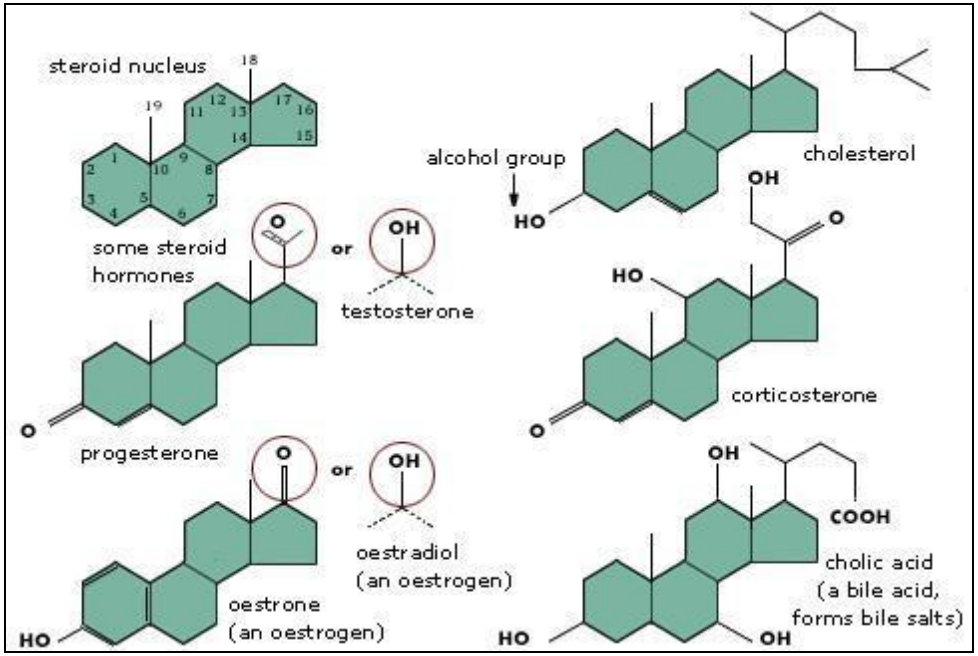
Lipoproteins	Site of Synthesis	Apo-lipoprotein	Function
Chylomicron	Intestine	B48 C-II, E, D	Transport of dietary lipids around the body
VLDL	Liver	B100 C-II, E, D	Transport of endogenous lipids (TAG) around the body
IDL	Plasma	B100 C-II, E	Transport of endogenous lipids (TAG) around the body
LDL	Plasma	B100	Transport of Cholesterol around body
HDL	Intestine and Liver	A-I Gives C-II, E, D	Transport Cholesterol from peripheral tissues to liver

CLASSIFICATION OF LIPOPROTEIN

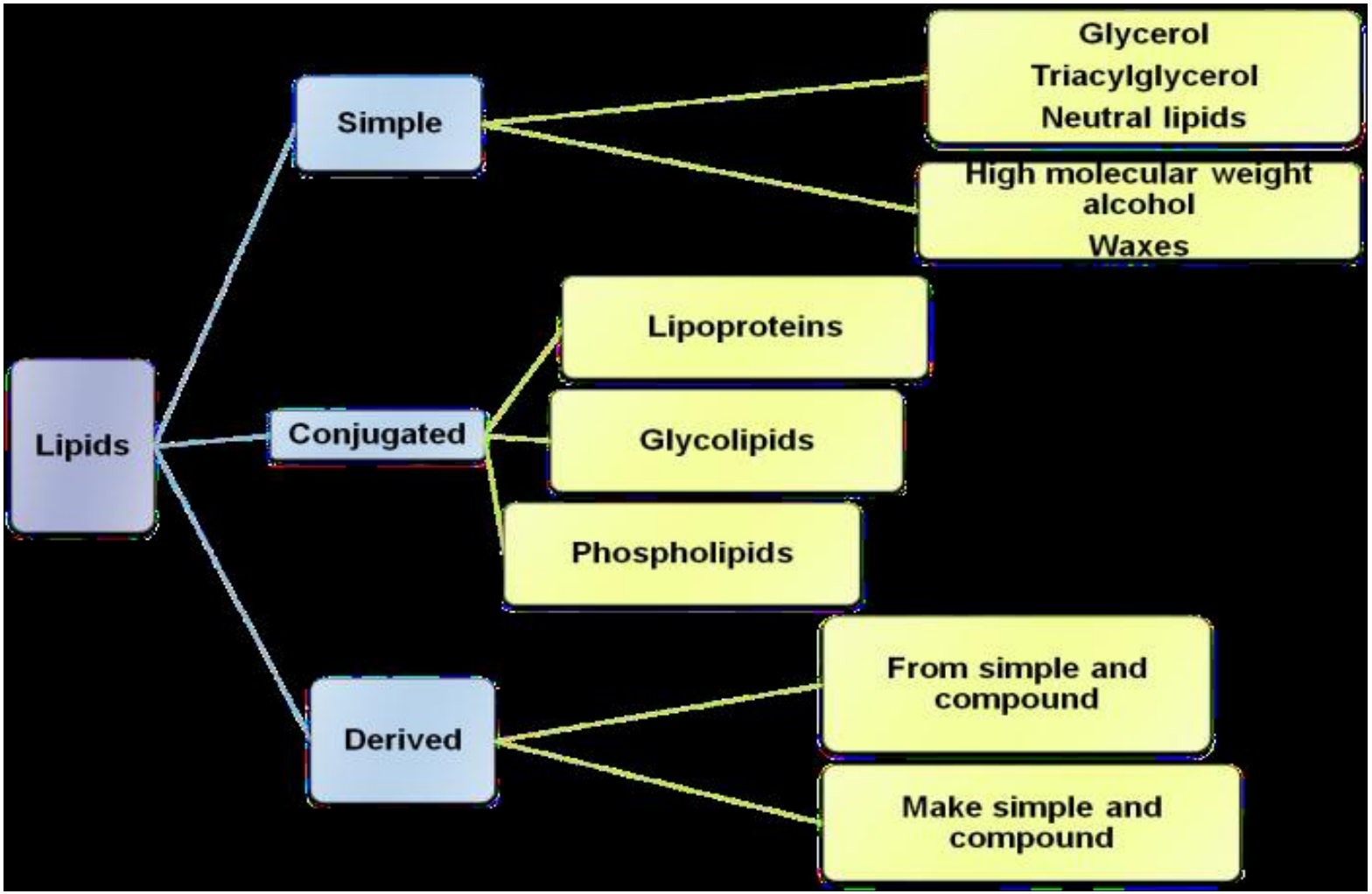
Lipo-proteins	Density	Triacyl-glycerol %	Cholesterol, Choleteryl Ester %	Phospho-lipids %	Protein %
Chylomicron	≤ 0.95	90	5	3	2
VLDL	1.00 – 1.05	60	20	15	5
LDL	1.10 – 1.15	8	50	22	20
HDL	≥ 1.20	5	25	30	40

DERIVED LIPIDS

- Molecules/group that are made from a lipid
  - Fatty acid
  - Glycerol
  - Sterol
  - Cholesterol
  - Sex hormones
  - Ketone bodies
  - Bile acids/salts
  - Fat soluble Vitamins
  - A,D,E and K.



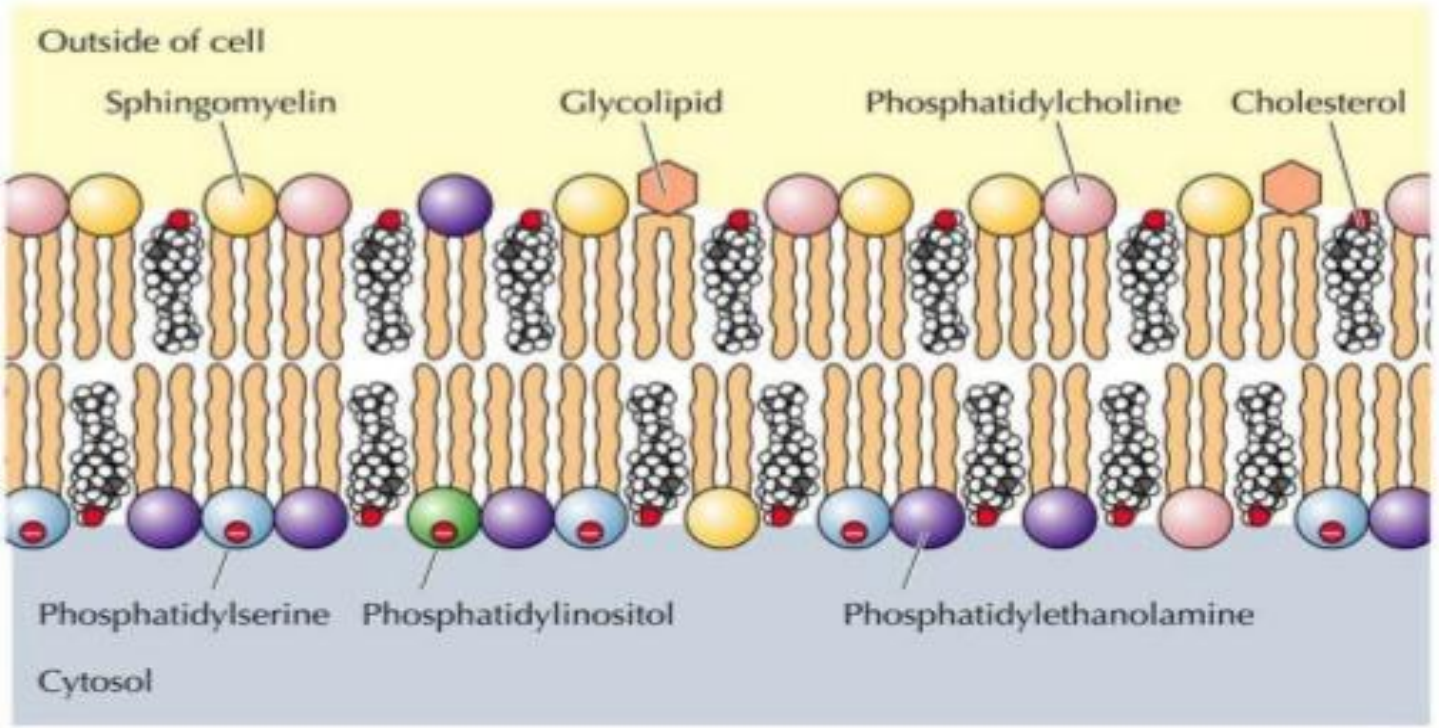
CLASSIFICATION OF LIPIDS



EXAMPLES OF LIPIDS

Simple	Compound or Conjugated or Complex	Derived
Triacylglycerol	Lipoproteins	Cholesterol
Ceramide	Glycolipids (Cerebroside, Ganglioside)	Retinol
Waxes	Phospholipids	Cholcalciferol
	Sphingolipids (Glyco and Phospho)	Ketone bodies
	Sulpholipids	Fatty acids
		Steroid hormones
		Ecosanoids Prostaglandins

Lipid Bilayer



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Clinical Significance of Lipids

- Following diseases are associated with abnormal chemistry or metabolism of lipids
  - Obesity
  - Atherosclerosis
  - DiabetesMellitus
  - Hyperlipoproteinemia
  - Fatty liver
  - Lipid storage diseases

REFERENCES

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  5th edition

  Ch 16, fatty acid & triacylglycerol metabolism

  Pg 181-182
- Harper's illustrated biochemistry

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Thank You